Formula analysis (Phase 2): repeated-n-gram coverage and oral-formulaic measures on live tennis commentary

Generated by `analysis/formulas/make_report.py` from `analysis/formulas/results/`; rerun everything with `bash analysis/formulas/run_all.sh`. The design was pre-specified in `analysis/formulas/plan.md` (committed before any result was computed); this is **revision 1**, made after the adversarial review `review/critic_analysis_v1.md`; every change is logged in plan.md, addendum 2, and summarised in section 10. Brackets are 95% intervals. Confirmatory tests are C1-C5 (2019 final) and their replication R1, R3-R5 (2023 final); everything else is exploratory, and analyses added after results were seen are labelled post hoc.

Terminology. The measure computed here is **repeated-n-gram coverage**: the share of word tokens that lie inside an exact word n-gram repeated in at least two utterances of an identification set. It is a rate of lexical repetition. It is not Parry's formula (an expression regularly used under the same metrical conditions to express a given essential idea), and it is not Duggan's "formulaic density", which counts repeated verse units (hemistichs) in a metrical text; tennis commentary has no metre, so no metrical condition can be imposed. "Formulaic density" appears below only in quotation marks, when the measure is related to Duggan's. To let the reader see how much of the repetition is short, function-word-heavy or numeric, every main design is also reported under a stricter family of definitions (section 1).

0. Data

corpus	role	kind of text	utterances	tokens (this tokeniser)
`tv_2019wimF`	MAIN, in-sample	TV commentary, raw ASR	318	9791
`tv_2023wimF`	HELD-OUT	TV commentary, raw ASR	245	7216
18 `tv_pool_*`	reference pool (identification)	TV commentary, raw ASR	2930	103675
`text_cornell`	written contrast	written live text (edited prose, one outlet)	3962	178770
press answers	spoken non-commentary baseline	player answers, edited stenographic transcript	81974	5435611

Utterance = one rally clip's de-duplicated, corrected ASR text (`text_corrected`; empty clips dropped), one live-text update, or one player answer. Mean tokens per utterance: tv_pool_all 35.4, tv_2019wimF 30.8, tv_2023wimF 29.5, text_cornell 45.1, press_answers 66.3. TV matches are dated 2019-2025 (match ids). The press answers (Cornell release, 358 interviewees, record dates 2007-2015) are human transcripts of a non-live genre; `corpus/SOURCES.md` lists them as DON'T USE *as commentary*; they are used only as a baseline (FOR_HUMAN.md). The Cornell live-text records carry no dates.

Key results

Table K. Repeated-n-gram coverage of the 2019 final under each definition (% of tokens [bootstrap 95% CI]; section 3, Table 3.1b gives the shuffled-word baselines and the excess over them).

definition	in-sample	split-half	pool -> 2019	2019 -> 2023
n >= 2, not stop-only (pre-registered (a))	44.2 [42.5, 46.3]	20.3 [18.9, 21.8]	55.0 [53.1, 57.0]	26.3 [24.7, 28.0]
n >= 3	19.4 [18.0, 20.9]	4.5 [3.7, 5.3]	24.4 [22.7, 26.1]	6.9 [5.8, 8.0]
n >= 4	6.0 [5.0, 7.0]	0.5 [0.2, 0.8]	7.1 [6.0, 8.3]	1.4 [0.9, 2.0]
n >= 2, not function-word/numeral-only	40.8 [39.0, 42.6]	18.4 [17.1, 19.8]	51.2 [49.4, 53.1]	24.0 [22.4, 25.5]
n >= 3, not function-word/numeral-only	18.7 [17.3, 20.2]	4.4 [3.6, 5.1]	23.8 [22.1, 25.4]	6.4 [5.3, 7.5]

- What the pre-registered measure counts: 56% of the covered tokens of the 2019 final have a bigram as their longest covering repeat (Table 2.2), and the most widespread items are `the first`, `a little`, `the match` (Table 2.4). Requiring n >= 3 cuts in-sample coverage from 44.2% to 19.4%; excluding n-grams made only of function words and numerals changes it much less (40.8%). Shuffling the words (a floor that keeps word frequencies and destroys syntax) leaves 16.2% in-sample coverage under the base definition but 0.5% for n >= 3, so the n >= 3 coverage lies almost wholly above that floor.
- Corpus contrasts at matched size (9,791 tokens, split-half, mean over 1000 subsamples): TV commentary pool 16.2%, Cornell written live text 35.3%, press answers 24.6% (n >= 3: 3.6%, 14.2%, 8.2%). These are contrasts between corpora that differ in matches, outlet, period, transcription (raw ASR vs edited text) and segmentation, not estimates of a medium effect (Table 3.2).
- C1/R1 (commentary vs press answers, held-out): **indeterminate pending a WER estimate**. TV 55.0% vs press 66.1% (difference interval [-13.9, -8.0] pp); but injected substitution noise brings the press value down to the TV value at e* = 0.122 (held-out; split-half e* = 0.142), while the only error estimate for the TV text is a lower bound of 22 visible errors per 1,000 words (Table 3.8, section 9.1).
- C2 (TV pool vs Cornell live text, split-half at matched size): corpus difference: 95% interval of differences excludes 0 (not attributable to medium); difference -19.1 pp, 95% interval over 1000 independent subsample pairs [-22.1, -16.3]. The pre-registered replicate-overlap "p" is 0.0020, its floor 2/(1000 + 1), and is not a calibrated test; the v1 Holm value 0.0498 (= 5 x 2/201) was a resolution artefact of R, not a borderline result.
- Timing tests (C3a, C3b, C4, C5 and replications): 0 of 8 rejected. With these sample sizes the tests detect, with 80% power at alpha = 0.05, only a coverage difference of about 6.4-7.2 pp between shortest and longest time terciles (C3a-R3b), a regime in which 30% (2019) or 42% (2023) of references take a tercile-specific form (C4/R4), or a length-time Spearman rho of 0.17 (C5; R5 0.22) (Table 4.3). Smaller effects would have gone undetected: the result is "not detected", not evidence of absence.
- Naming (commentary-only references, after removing 31 (2019) and 31 (2023) references inside umpire/Hawk-Eye patterns): the bare surname is 67-85% of each player's references and descriptive epithets 4-7% (sections 5-6).

1. Definitions (exact)

Tokens. NFKC; curly quotes to `'`; hyphens, dashes and `/` to space; lower-case; token = `[a-z0-9]+(?:'[a-z0-9]+)*` (clitics stay attached: `it's`, `federer's`); other punctuation dropped; digits kept verbatim (`15`, `40`); number words not normalised. N-grams never cross an utterance boundary but do cross punctuation inside an utterance (WhisperX punctuation is unreliable). STOP = a frozen list of 182 function words (`common.py`); digits and player names are never STOP.

(a) Repeated n-gram ("formula" in the plan). An n-gram (2 <= n <= 12) is a repeated n-gram of an identification set I if it occurs in at least m = 2 distinct utterances of I and is not stop-only. **Repeated-n-gram coverage** of a measurement set M = share of M's tokens lying inside at least one occurrence (in M) of a repeated n-gram of I. In-sample (I = M) a token is covered if its n-gram recurs elsewhere in the same text, the operational analogue of Duggan's counting of repeated units (the wording of Duggan's own criterion is [unverified]: only the bibliographic record was fetched, see below). Held-out (I and M disjoint) it is the share of M that reuses n-grams already repeated in I.

Stricter coverage family (revision 1, exploratory). Same I, M and repeated-n-gram inventory; a token counts only if it lies inside a qualifying repeated n-gram: `base` = n >= 2, not stop-only (pre-registered (a)); `n3` = n >= 3; `n4` = n >= 4; `content` = n >= 2, not function-word/numeral-only; `content_n3` = n >= 3, not function-word/numeral-only. Numerals = digit strings and the 45 words of `common.NUMERAL_WORDS` (tennis score words `love fifteen thirty forty deuce`, English cardinals and the ordinals `first-twelfth`), so `the first` and `40 15` are function/numeral-only, while `the first set` and `a little` are not.

(b) System. A frame is an n-gram (2 <= n <= 6) with exactly one slot. Slot types: `<NAME>` (a first name or surname of any of the players of the 20 TennisVL matches, 44 name

tokens, possessive stripped), <NUM> (a digit string or love fifteen thirty forty deuce), <_> (any token). A frame is a system of I if it occurs >= 3 times in >= 3 utterances of I with >= 2 distinct fillers, has >= 1 non-STOP fixed token, and, if the slot is <_>, the slot is internal (fixed tokens on both sides). (a)+(b) coverage = share of tokens covered by a repeated n-gram or by an occurrence of a system of I (any filler of the right type).

Shuffled-word baselines. In-sample and split-half (D7, pre-registered): all tokens of the 2019 stream permuted over positions (utterance lengths kept), then identification and measurement repeated (R = 200). Held-out (revision 1): the tokens of M permuted the same way while I's inventory is kept. Excess = coverage minus the mean shuffled coverage; its interval pairs bootstrap draws of the coverage with shuffled replicates at random. The baseline keeps word frequencies and destroys syntax, so it is a floor, not a syntax-preserving null.

Designs. D1 in-sample (I = M). D2 split-half: I = odd clips, M = even clips and vice versa (pooled; CI resamples within halves). D3 held-out across matches: I = the 18 pool matches, M = the 2019 final; I = the 2019 final, M = the 2023 final. D4 per stream (leave-one-stream-out identification; and small subsamples). D5 per corpus at matched size: (i) in-sample and split-half in random subsamples of the 2019 size; (ii) held-out with I and M from disjoint groups (pre-registered); (ii-b, post hoc) held-out with M concentrated in as few groups as possible; (iii) in-sample at pool size. D6 per context. D7 shuffled-word baseline.

Contexts. phase is the corpus's regex tag (between_points = the whole clip text is a score call; changeover = a cue word); it is defined from the text itself, so its contrast is partly definitional. dead_time_before_s = PBP time from the end of the previous rally to this point's first serve. time_after_s = dead_time_before_s of the next point (or the gap to the second serve for a first-serve-fault clip), used because the clip transcript runs past the clip end (corpus/README section 4). Terciles are cut per stream: 2019 dead time before 24.2 / 35.9 s, time after 22.8 / 32.7 s; 2023 32.1 / 44.4 s and 28.8 / 42.4 s. Score situation of the aligned PBP point: match point > set point > break point > other tie-break point > deuce/advantage > other.

Umpire-type patterns (revision 1). Two pattern sets, both in the code: (i) common.official_mask_v2 (= the pre-registered S5 patterns plus mr <name>, advantage <name>, game (and) set (and match) <name>, game and <ordinal> set <name>, <name>... (is) challenging (the call), <name>... has <x> challenge(s) remaining/left, bare thank you (+ players/please/all), point-score calls <score> <score|all> with score = 0 15 30 40 love fifteen thirty forty and the ASR confusions 13 14 50, and deuce), used for the top-10 official flag and the S5b sensitivity; (ii) refexpr.umpire_rule on the word sequence around a player reference: game_name before = (?:^|)game$; game_set_name before = (?:^|)game (?:and)?(?:\w+)?set(?: and match)?$; advantage_name before = (?:^|)advantage$; leads_games after = ^(?:leads?|leading) (?:by)?(?:by)?\w+ (?:games?|gains?) to\b; challenging after = ^(?:is)?(?:challenging|charging|having a call)\b; challenges_remaining after = ^has (?:\w+) challenges?(?:remaining|left)\b; and every mr <surname>. Score calls are said by the umpire and by the commentators; the transcript does not tell them apart.

Uncertainty. Coverage: percentile bootstrap over utterances of M (B = 2000), with I's inventory held fixed, so the interval covers sampling of the measured text, not uncertainty in identification. Subsample designs report the mean and 2.5-97.5% range over replicates. Tests: section 4.

Reference for Duggan. Duggan, Joseph J. 1973. The Song of Roland: Formulaic Style and Poetic Craft. Berkeley: University of California Press. 226 pp. (Publications of the Center for Medieval and Renaissance Studies, UCLA, 6). Status: **verified (bibliographic only)**. Record: Open Library edition record https://openlibrary.org/books/OL5224003M.json (LCCN 75186101, OCLC 629242, ISBN 0520022017); Crossref record of the reissue https://api.crossref.org/works/10.2307/jj.8085370 (chapter "Formulaic Language and Mode of Creation", pp. 16-62, doi:10.2307/jj.8085370.4); Crossref record of Friedman's review, Speculum 51(1) 1976, pp. 126-129, doi:10.2307/2851011. Confirms: author, title, subtitle, publisher, place, year, pages, series; the book was not read, so the wording and operational details of Duggan's formula criterion are [unverified].

2. Repeated n-grams (a) and systems (b) in the 2019 final

1194 repeated n-gram types (n = 2..12, stop-only n-grams excluded; 1503 without the filter; 1113 after also excluding n-grams made only of function words and numerals). In-sample (a) coverage 44.2% [42.5, 46.3]. 919 of the 1194 types also occur in at least one of the 18 other matches, and 467 in the 2023 final.

Table 2.1. Repeated n-grams by length n (in-sample, 2019). Coverage = share of all 2019 tokens covered by repeats of exactly this length.

n	filter	types	occurrences	coverage %	types in >= 1 / >= 3 / >= 9 pool matches	types in 2023
2	stop_filtered	745	2501	42.1	640 / 517 / 218	353
2	unfiltered	1020	3750	58.5	906 / 766 / 383	542
3	stop_filtered	314	793	19.3	221 / 147 / 48	96
3	unfiltered	346	868	20.9	248 / 168 / 56	108
4	stop_filtered	90	196	5.9	45 / 25 / 5	15
4	unfiltered	92	200	6.0	47 / 26 / 5	15
5	stop_filtered	31	65	2.2	9 / 7 / 1	3
5	unfiltered	31	65	2.2	9 / 7 / 1	3
6	stop_filtered	11	23	1.1	3 / 3 / 0	0
6	unfiltered	11	23	1.1	3 / 3 / 0	0
7	stop_filtered	3	6	0.4	1 / 1 / 0	0
7	unfiltered	3	6	0.4	1 / 1 / 0	0

Table 2.2. Maximal match (each token assigned the length of the longest repeat covering it; 0 = uncovered).

longest repeat covering the token	tokens	%
0	5459	55.8
2	2430	24.8
3	1318	13.5
4	367	3.7
5	109	1.1
6	66	0.7
7	42	0.4

Table 2.3. Within vs across matches: 2019 repeated n-gram types by the number of other matches (pool streams, each a different match and possibly a different broadcaster) in which they occur.

pool matches attesting the n-gram	2019 types
0	275
1-2	219
3-8	428
9-18	272

Table 2.4. The 25 most widespread repeated n-grams (by distinct utterances; full list with excerpts in `results/formulas_2019.tsv`). `thank you` (rank 6) is mostly the umpire's call to the crowd (`thank you players`, `thank you please`), counted here as pre-registered; it is masked in S5b and flagged `official` in the top-10 lists (section 7).

rank	n-gram	n	utterances	occurrences	coverage %	pool matches
1	`the first`	2	24	28	0.57	18
2	`a little`	2	22	24	0.49	18
3	`the match`	2	17	17	0.35	15
4	`from djokovic`	2	16	16	0.33	4
5	`going to`	2	15	17	0.35	18
6	`thank you`	2	15	26	0.53	17
7	`you know`	2	15	18	0.37	16
8	`a little bit`	3	14	15	0.46	18
9	`little bit`	2	14	15	0.31	18

rank	n-gram	n	utterances	occurrences	coverage %	pool matches
10	novak djokovic	2	14	16	0.33	6
11	the forehand	2	13	15	0.31	16
12	the line	2	13	13	0.27	18
13	30 15	2	12	12	0.24	11
14	40 15	2	12	13	0.27	5
15	djokovic has	2	12	12	0.24	4
16	fifth set	2	12	12	0.24	5
17	i think	2	12	16	0.33	18
18	the net	2	12	12	0.24	16
19	to get	2	12	12	0.24	18
20	second set	2	11	11	0.22	18
21	the fifth	2	11	11	0.22	6
22	an hour	2	10	10	0.20	14
23	second serve	2	10	11	0.22	17
24	the backhand	2	10	10	0.20	15
25	the court	2	10	10	0.20	18

Systems. 231 systems: NUM 40, OPEN 150, NAME 41. By length: n=2 NAME 22, n=2 NUM 31, n=3 NAME 11, n=3 NUM 6, n=3 OPEN 108, n=4 NAME 5, n=4 NUM 3, n=4 OPEN 28, n=5 NAME 2, n=5 OPEN 10, n=6 NAME 1, n=6 OPEN 4. In 84 of the 231 systems the commonest filler takes at least two thirds of the occurrences (column 'top filler %'); such a frame barely alternates (e.g. `<NAME> djokovic` is mostly the full name `novak djokovic`), so it is a system only by the letter of criterion (b).

Table 2.5. The 25 most widespread systems (`<_>` open slot; full list in `results/systems_2019.tsv`). 'beyond (a)' = share of all tokens covered by the frame but by no repeated n-gram.

rank	frame	slot	occurrences	utterances	fillers (top)	top filler %	coverage %	beyond (a) %	pool matches
1	<NUM> 15	NUM	40	36	40:13; 30:12; love:7; 13:3; 14:2	32	0.82	0.05	16
2	15 <NUM>	NUM	30	29	love:8; 0:7; 30:6; 13:5; 40:2	27	0.61	0.00	16
3	<NUM> 30	NUM	26	24	40:10; 15:6; 14:5; 0:3; love:2	38	0.53	0.00	16
4	40 <NUM>	NUM	28	22	15:13; 30:10; love:4; 40:1	46	0.57	0.02	17
5	the <_> set	OPEN	27	21	first:10; fifth:7; second:6; fourth:2; third:1	37	0.83	0.00	17
6	<NAME> djokovic	NAME	21	19	novak:16; federer:3; roger:1; djokovic:1	76	0.43	0.03	6
7	30 <NUM>	NUM	19	19	15:12; love:4; 40:2; 0:1	63	0.39	0.02	16
8	<NUM> love	NUM	19	15	15:8; 30:4; 40:4; 50:1; thirty:1	42	0.39	0.06	13
9	novak <NAME>	NAME	17	15	djokovic:16; djokovic's:1	94	0.35	0.01	7
10	<NUM> 0	NUM	16	15	15:7; 50:3; 1:3; 30:1; 12:1	44	0.33	0.05	14

rank	frame	slot	occurrences	utterances	fillers (top)	top filler %	coverage %	beyond (a) %	pool matches
11	`<NUM> minutes`	NUM	14	12	10:3; 48:2; 20:1; 58:1; 13:1	21	0.29	0.16	15
12	`djokovic <_> the`	OPEN	11	11	finding:2; had:2; down:1; certainly:1; perhaps:1	18	0.34	0.10	4
13	`mr <NAME>`	NAME	11	11	federer:6; djokovic:5	55	0.22	0.00	2
14	`federer <_> to`	OPEN	10	10	looking:3; going:1; trying:1; able:1; yet:1	30	0.31	0.09	0
15	`love <NUM>`	NUM	11	9	15:7; 30:2; 40:1; forty:1	64	0.22	0.02	16
16	`by <_> games to`	OPEN	9	9	two:3; four:2; six:1; five:1; three:1	33	0.37	0.04	7
17	`in the <_> set`	OPEN	9	9	fifth:4; first:2; second:2; final:1	44	0.37	0.00	15
18	`by <_> games`	OPEN	9	9	two:3; four:2; six:1; five:1; three:1	33	0.28	0.04	7
19	`in <_> fifth`	OPEN	9	9	the:6; this:3	67	0.28	0.00	4
20	`<NAME> federer`	NAME	9	9	roger:8; federer:1	89	0.18	0.00	4
21	`game <NAME>`	NAME	9	9	djokovic:7; federer:2	78	0.18	0.00	9
22	`roger <NAME>`	NAME	9	9	federer:8; djokovic:1	89	0.18	0.02	4
23	`14 <NUM>`	NUM	9	8	30:5; 15:2; 13:2	56	0.18	0.00	8
24	`djokovic <_> to`	OPEN	8	8	trying:2; able:2; needs:1; serving:1; had:1	25	0.24	0.05	3
25	`federer <_> the`	OPEN	8	8	with:1; got:1; resisting:1; gets:1; was:1	12	0.24	0.06	2

Table 2.6. Most widespread repeated n-grams of n >= 3 in each corpus
(exploratory; each corpus identified on itself, whole corpus; utterance counts and rate per 1,000 tokens; n-grams contained in a longer listed one with the same count dropped).

rank	2019 final	TV pool (18 matches)	Cornell live text
1	`a little bit` 14 (1.43)	`a little bit` 139 (1.34)	`into the net` 366 (2.05)
2	`the first set` 8 (0.82)	`a lot of` 93 (0.90)	`a couple of` 231 (1.29)
3	`of the match` 7 (0.71)	`of the court` 68 (0.66)	`of the match` 229 (1.28)
4	`the fifth set` 7 (0.71)	`a couple of` 66 (0.64)	`down the line` 223 (1.25)
5	`at the moment` 6 (0.61)	`going to be` 62 (0.60)	`the world number` 221 (1.24)
6	`down the line` 6 (0.61)	`down the line` 54 (0.52)	`the next point` 206 (1.15)
7	`going to be` 6 (0.61)	`at the moment` 51 (0.49)	`the game with` 202 (1.13)
8	`in the fifth` 6 (0.61)	`one of the` 51 (0.49)	`at the net` 200 (1.12)
9	`the second set` 6 (0.61)	`the second set` 43 (0.41)	`a double fault` 198 (1.11)
10	`and a half` 5 (0.51)	`in the second` 37 (0.36)	`with a forehand` 191 (1.07)
11	`at the end` 5 (0.51)	`the first serve` 37 (0.36)	`with an ace` 166 (0.93)

rank	2019 final	TV pool (18 matches)	Cornell live text
12	`at this point` 5 (0.51)	`is going to` 36 (0.35)	`a forehand winner` 153 (0.86)
13	`is going to` 5 (0.51)	`a bit of` 35 (0.34)	`of the court` 149 (0.83)
14	`the first point` 5 (0.51)	`little bit of` 34 (0.33)	`two break points` 148 (0.83)
15	`the semi final` 5 (0.51)	`in this match` 33 (0.32)	`the second set` 145 (0.81)

ASR in score calls: of 195 filler occurrences in two-token score-call frames, 29 are not possible score words (`14` 9, `13` 8, `50` 4, `1` 3, `10` 1), i.e. 'forty' heard as `14`, 'thirty' as `13`, 'fifteen' as `50`. These split exact score-call repeats but are absorbed by `<NUM>` systems.

3. Repeated-n-gram coverage

Table 3.1. Main coverage, pre-registered definitions (TV commentary, finals).

design	identified on	measured on	tokens	(a) %	(a)+(b) %
D1_in_sample	tv_2019wimF	tv_2019wimF	9791	44.2 [42.5, 46.3]	48.3 [46.4, 50.5]
D1_in_sample	tv_2023wimF	tv_2023wimF	7216	43.2 [41.3, 45.1]	46.8 [44.8, 48.8]
D2_split_half	2019 odd clips	2019 even clips	4605	21.0 [19.3, 22.9]	25.1 [23.1, 27.2]
D2_split_half	2019 even clips	2019 odd clips	5186	19.6 [17.5, 22.1]	21.8 [19.6, 24.6]
D2_split_half	2019 halves	2019 other half (pooled)	9791	20.3 [18.9, 21.8]	23.3 [21.8, 25.1]
D2_split_half	2023 halves	2023 other half (pooled)	7216	18.4 [16.7, 20.0]	20.6 [18.9, 22.5]
D3_held_out	pool (18 matches)	tv_2019wimF	9791	55.0 [53.1, 57.0]	63.9 [61.9, 65.9]
D3_held_out	tv_2019wimF	tv_2023wimF	7216	26.3 [24.7, 28.0]	29.9 [28.2, 31.7]
D3_held_out	pool (18 matches)	tv_2023wimF	7216	59.5 [57.6, 61.5]	67.6 [65.6, 69.5]

In-sample and held-out figures differ as the sizes of I lead one to expect: the 2019 final reuses 44.2% of its tokens within itself, 20.3% when the repeats must come from the other half of the match, and 55.0% when they come from 18 other matches (103,675 tokens, about 11 times the final). Coverage therefore depends strongly on the size of I; compare only like with like.

Table 3.1b. Coverage family with shuffled-word baselines (revision 1, exploratory). Coverage [bootstrap 95% CI]; shuffled = mean [2.5-97.5%] over shuffled replicates; excess = coverage minus shuffled mean [interval]. Held-out baselines shuffle M only.

design	definition	coverage %	shuffled %	excess (pp)
in-sample	base	44.2 [42.5, 46.3]	16.2 [15.2, 17.3]	28.1 [26.0, 30.3]
in-sample	n3	19.4 [18.0, 20.9]	0.5 [0.2, 0.8]	18.9 [17.4, 20.4]
in-sample	n4	6.0 [5.0, 7.0]	0.0 [0.0, 0.1]	6.0 [5.0, 7.1]
in-sample	content	40.8 [39.0, 42.6]	13.0 [12.0, 13.9]	27.8 [25.7, 29.8]
in-sample	content_n3	18.7 [17.3, 20.2]	0.4 [0.1, 0.7]	18.3 [16.9, 19.8]
split-half	base	20.3 [18.9, 21.8]	2.8 [2.2, 3.5]	17.5 [15.9, 19.1]
split-half	n3	4.5 [3.7, 5.3]	0.0 [0.0, 0.0]	4.4 [3.7, 5.3]
split-half	n4	0.5 [0.2, 0.8]	0.0 [0.0, 0.0]	0.5 [0.2, 0.8]
split-half	content	18.4 [17.1, 19.8]	2.2 [1.7, 2.8]	16.2 [14.8, 17.8]
split-half	content_n3	4.4 [3.6, 5.1]	0.0 [0.0, 0.0]	4.3 [3.6, 5.1]
pool -> 2019	base	55.0 [53.1, 57.0]	17.2 [16.3, 18.1]	37.9 [35.8, 40.0]
pool -> 2019	n3	24.4 [22.7, 26.1]	0.5 [0.3, 0.7]	23.9 [22.2, 25.6]
pool -> 2019	n4	7.1 [6.0, 8.3]	0.0 [0.0, 0.0]	7.1 [6.0, 8.3]
pool -> 2019	content	51.2 [49.4, 53.1]	14.5 [13.7, 15.2]	36.7 [34.7, 38.8]
pool -> 2019	content_n3	23.8 [22.1, 25.4]	0.4 [0.2, 0.6]	23.4 [21.7, 25.0]
2019 -> 2023	base	26.3 [24.7, 28.0]	4.6 [4.0, 5.2]	21.7 [20.0, 23.5]
2019 -> 2023	n3	6.9 [5.8, 8.0]	0.0 [0.0, 0.1]	6.8 [5.7, 8.0]
2019 -> 2023	n4	1.4 [0.9, 2.0]	0.0 [0.0, 0.0]	1.4 [0.9, 2.0]
2019 -> 2023	content	24.0 [22.4, 25.5]	4.0 [3.5, 4.6]	20.0 [18.3, 21.6]
2019 -> 2023	content_n3	6.4 [5.3, 7.5]	0.0 [0.0, 0.1]	6.4 [5.3, 7.5]
pool -> 2023	base	59.5 [57.6, 61.5]	18.6 [17.6, 19.6]	40.8 [38.7, 43.0]
pool -> 2023	n3	27.3 [25.2, 29.3]	0.6 [0.3, 0.8]	26.8 [24.7, 28.8]
pool -> 2023	n4	8.0 [6.8, 9.4]	0.0 [0.0, 0.1]	8.0 [6.8, 9.4]
pool -> 2023	content	55.7 [53.8, 57.6]	16.2 [15.2, 17.2]	39.5 [37.3, 41.6]

design	definition	coverage %	shuffled %	excess (pp)
pool -> 2023	content_n3	26.2 [24.2, 28.2]	0.5 [0.2, 0.8]	25.7 [23.7, 27.8]
2023 in-sample	base	43.2 [41.3, 45.1]	-	-
2023 in-sample	n3	20.0 [18.1, 21.9]	-	-
2023 in-sample	n4	7.3 [5.7, 9.0]	-	-
2023 in-sample	content	40.3 [38.3, 42.2]	-	-
2023 in-sample	content_n3	19.1 [17.2, 21.0]	-	-
2023 split-half	base	18.4 [16.7, 20.0]	-	-
2023 split-half	n3	4.2 [3.3, 5.3]	-	-
2023 split-half	n4	1.6 [1.0, 2.3]	-	-
2023 split-half	content	16.5 [14.9, 18.0]	-	-
2023 split-half	content_n3	3.6 [2.8, 4.6]	-	-

Table 3.2. Corpus contrasts at matched size (each corpus identified and measured on itself; mean and 2.5-97.5% over replicates). TV pool, Cornell and press differ in matches, outlet, period, transcription (raw ASR vs edited prose vs edited stenography) and segmentation (rally-clip windows that cut sentences vs whole updates vs whole answers); the matched-size design equalises tokens only. Held-out columns give the realised size of I: the pre-registered rule (I = 100,000 tokens from groups disjoint from M) could not be met for Cornell, because excluding every player pair touched by M removes most of the corpus (DEVIATION, plan.md addendum 2).

corpus	tokens measured	in-sample (a) %, R = 1000	split-half (a) %, R = 1000	held-out (a) %, D5(ii) as pre-registered, R = 50	held-out (a) %, M in few groups, D5(ii-b) post hoc, R = 50
shuffled words, 2019 (D7)	9,791	16.2 [15.2, 17.3]	2.8 [2.2, 3.5]	-	-
2019 final (single match)	9,791	44.2 [42.5, 46.3]	20.3 [18.9, 21.8]	55.0 [53.1, 57.0] (pool -> 2019, I = 103,675)	-
TV commentary, pool subsample (18 matches)	9,791	38.9 [37.0, 40.8]	16.2 [14.5, 17.8]	see D3/D4	-
Cornell live text (written)	9,791	60.7 [58.6, 62.9]	35.3 [32.9, 37.7]	70.9 [67.8, 72.8]; I = 65,444 (56,574-76,850); M from 138 groups	76.0 [71.7, 79.9]; I = 103,706; M from 29 groups (7-42)
press answers (spoken, not commentary)	9,791	48.6 [46.1, 51.0]	24.6 [22.3, 26.9]	66.1 [64.1, 68.1]; I = 103,704 (103,675-103,860); M from 70 groups	66.4 [60.9, 72.3]; I = 103,710; M from 2 groups (1-9)

Cornell size correction (B2) and the concentrated-M check: Cornell 70.9% (D5(ii)) vs 76.0% (D5(ii-b)), change 5.1 pp; relative to TV pool -> 2019 (55.0%) the direction is unchanged; press 66.1% (D5(ii)) vs 66.4% (D5(ii-b)), change 0.3 pp; relative to TV pool -> 2019 (55.0%) the direction is unchanged. D5(ii-b) changes two things at once (I reaches the pool size, and M comes from few groups: for press usually one interviewee, the closest analogue of TV's one-match M), so the change is not attributable to either alone.

Table 3.2b. Corpus contrasts under the stricter definitions (split-half at the 2019 size, mean [2.5-97.5%] over 1000 subsamples; in-sample n >= 3 for reference).

corpus	split-half base	split-half n >= 3	split-half n >= 4	split-half content	split-half content n >= 3	in-sample n >= 3
TV commentary, pool subsample (18 matches)	16.2 [14.5, 17.8]	3.6 [2.6, 4.6]	0.4 [0.1, 0.9]	15.1 [13.5, 16.7]	3.4 [2.4, 4.4]	15.6 [13.8, 17.4]
Cornell live text (written)	35.3 [32.9, 37.7]	14.2 [12.2, 16.2]	3.5 [2.3, 4.9]	31.9 [29.5, 34.1]	13.7 [11.6, 15.6]	35.8 [33.2, 38.3]
press answers (spoken, not commentary)	24.6 [22.3, 26.9]	8.2 [6.6, 10.0]	1.0 [0.4, 1.8]	24.0 [21.8, 26.3]	8.0 [6.5, 9.7]	25.3 [22.9, 27.8]

Table 3.2c. Held-out coverage family across corpora (TV: pool -> 2019 final, bootstrap CI; Cornell and press: D5(ii) and D5(ii-b), mean [2.5-97.5%] over replicates).

corpus, design	base	n3	n4	content	content_n3
TV pool -> 2019 (I = 103,675)	55.0 [53.1, 57.0]	24.4 [22.7, 26.1]	7.1 [6.0, 8.3]	51.2 [49.4, 53.1]	23.8 [22.1, 25.4]
Cornell live text (written), D5(ii)	70.9 [67.8, 72.8]	45.8 [42.6, 48.5]	22.4 [20.0, 25.0]	67.8 [64.9, 69.6]	43.8 [40.6, 46.4]
press answers (spoken, not commentary), D5(ii)	66.1 [64.1, 68.1]	40.2 [37.8, 42.1]	15.3 [13.8, 16.9]	65.0 [62.8, 66.9]	39.6 [37.3, 41.6]
Cornell live text (written), D5(ii-b)	76.0 [71.7, 79.9]	52.2 [46.2, 56.6]	27.5 [22.8, 31.3]	73.3 [69.2, 77.5]	50.2 [44.5, 54.6]
press answers (spoken, not commentary), D5(ii-b)	66.4 [60.9, 72.3]	40.3 [33.9, 46.9]	15.5 [11.6, 20.4]	65.3 [59.9, 71.5]	39.8 [33.1, 46.1]

Table 3.3. (a)+(b) at 2019 size (R = 200), and (a) in-sample at pool size (103,675 tokens, R = 20).

corpus	in-sample (a)+(b) %	split-half (a)+(b) %	in-sample (a) % at pool size	in-sample n >= 3 % at pool size
TV commentary, pool subsample (18 matches)	42.4 [40.3, 44.5]	17.7 [16.1, 19.2]	69.8 [69.8, 69.8]	41.7 [41.7, 41.7]
Cornell live text (written)	66.5 [64.4, 68.5]	41.0 [38.4, 43.6]	85.0 [84.7, 85.2]	65.8 [65.5, 66.2]
press answers (spoken, not commentary)	52.8 [50.3, 55.3]	26.3 [23.9, 28.8]	76.9 [76.4, 77.4]	55.0 [54.2, 55.8]

At pool size the TV 'subsample' is the whole pool, so its replicates are identical. Written live text is identified and measured only on itself; nothing is pooled across corpora.

Table 3.4. Per stream (each stream = one match with one unnamed TV broadcaster; broadcaster, match and commentators are confounded). Leave-one-out: I = the pool without this stream (for the finals: the whole pool). 1,500-token columns: mean over 200 random subsamples.

stream	tokens	leave-one-out held-out (a) %	in-sample (a) %, 1,500 tokens	split-half (a) %, 1,500 tokens
tv_2019wimF	9791	55.0 [53.2, 57.1]	19.6 [15.0, 24.4]	3.0 [1.1, 5.6]
tv_2023wimF	7216	59.5 [57.6, 61.3]	20.2 [15.2, 25.1]	3.2 [1.2, 5.7]
text_cornell	-	-	33.1 [28.0, 38.1]	8.8 [5.3, 12.4]
press_answers	-	-	23.7 [17.8, 29.4]	7.2 [3.8, 11.0]
pool: 20190126-W-Australian_Open-F-Naomi_Osaka-Petra_Kvitova	6527	51.8 [49.4, 54.2]	20.5 [16.7, 25.5]	3.4 [1.3, 6.2]
pool: 20200202-M-Australian_Open-F-Novak_Djokovic-Dominic_Th	8371	56.7 [55.0, 58.4]	20.4 [15.2, 24.8]	4.4 [2.1, 7.0]
pool: 20201010-W-Roland_Garros-F-Iga_Swiatek-Sofia_Kenin	1597	62.2 [58.1, 66.5]	24.8 [23.3, 25.6]	6.2 [4.3, 8.0]
pool: 20210220-W-Australian_Open-F-Naomi_Osaka-Jennifer_Brad	3276	54.2 [51.0, 57.3]	23.9 [19.8, 27.7]	5.5 [3.3, 7.6]
pool: 20210613-M-Roland_Garros-F-Stefanos_Tsitsipas-Novak_Dj	7837	55.8 [54.0, 57.6]	21.8 [18.3, 25.6]	3.5 [1.2, 6.3]
pool: 20210911-W-US_Open-F-Emma_Raducanu-Leylah_Fernandez	3498	50.5 [47.6, 53.7]	19.3 [15.6, 23.0]	3.5 [1.7, 5.3]
pool: 20220130-M-Australian_Open-F-Rafael_Nadal-Daniil_Medve	11505	56.8 [55.2, 58.4]	20.4 [15.8, 25.5]	4.2 [1.7, 7.1]
pool: 20220531-M-Roland_Garros-QF-Novak_Djokovic-Rafael_Nada	8046	57.0 [55.1, 58.7]	20.9 [15.8, 25.7]	3.1 [1.1, 5.4]
pool: 20220911-M-US_Open-F-Casper_Ruud-Carlos_Alcaraz	6231	53.9 [51.7, 56.1]	19.2 [15.0, 23.5]	2.8 [1.1, 4.8]
pool: 20230610-W-Roland_Garros-F-Iga_Swiatek-Karolina_Muchov	3800	60.1 [57.6, 62.5]	28.1 [24.3, 32.3]	6.6 [3.7, 9.7]
pool: 20230910-M-US_Open-F-Novak_Djokovic-Daniil_Medvedev	5992	56.8 [54.5, 59.0]	19.0 [15.3, 22.9]	3.0 [0.9, 5.6]
pool: 20240128-M-Australian_Open-F-Jannik_Sinner-Daniil_Medv	9256	54.4 [52.6, 56.2]	19.0 [15.2, 23.4]	3.3 [1.5, 5.5]

stream	tokens	leave-one-out held-out (a) %	in-sample (a) %, 1,500 tokens	split-half (a) %, 1,500 tokens
pool: 20240608-W-Roland_Garros-F-Iga_Swiatek-Jasmine_Paolini	2368	62.2 [59.5, 64.9]	27.1 [23.8, 30.9]	4.9 [3.0, 6.7]
pool: 20240713-W-Wimbledon-F-Jasmine_Paolini-Barbora_Krejcik	3825	54.1 [50.6, 57.2]	22.5 [17.9, 27.4]	3.6 [1.5, 5.6]
pool: 20240908-M-US_Open-F-Taylor_Fritz-Jannik_Sinner	5756	48.9 [46.5, 51.5]	15.9 [11.8, 19.5]	2.6 [1.0, 4.9]
pool: 20250124-M-Australian_Open-SF-Jannik_Sinner-Ben_Shelto	7113	56.2 [54.1, 58.1]	20.0 [15.8, 24.0]	3.5 [1.7, 5.7]
pool: 20250712-W-Wimbledon-F-Amanda_Anisimova-Iga_Swiatek	2592	48.2 [44.9, 51.2]	15.7 [13.1, 18.1]	2.0 [0.9, 3.4]
pool: 20250907-M-US_Open-F-Jannik_Sinner-Carlos_Alcaraz	6085	52.8 [50.6, 55.0]	17.1 [13.3, 21.7]	2.4 [0.9, 4.4]

Table 3.5. Per context, 2019 final ((a) coverage; in-sample = repeats from the whole 2019 stream; pool = repeats from the 18 other matches).

context	group	utterances	tokens	in-sample (a) %	pool (a) %	pool (a)+(b) %
phase tag (heuristic)	between_points	14	38	94.7 [84.2, 100.0]	94.7 [84.2, 100.0]	100.0 [100.0, 100.0]
phase tag (heuristic)	changeover	4	164	45.7 [36.9, 64.0]	62.8 [50.7, 83.3]	76.2 [68.2, 95.7]
phase tag (heuristic)	clip	300	9589	44.0 [42.2, 46.0]	54.7 [52.9, 56.8]	63.6 [61.5, 65.6]
dead time before (tercile)	NA	36	1056	40.1 [34.7, 47.0]	51.9 [44.8, 60.0]	59.8 [51.3, 69.2]
dead time before (tercile)	T1	94	3021	45.1 [42.5, 48.1]	55.4 [52.1, 58.7]	65.1 [62.0, 68.1]
dead time before (tercile)	T2	94	2835	44.4 [40.8, 49.1]	55.8 [52.2, 59.9]	64.8 [61.2, 68.9]
dead time before (tercile)	T3	94	2879	44.7 [41.6, 48.1]	55.1 [52.3, 58.2]	63.3 [60.2, 66.4]
time after (tercile)	NA	12	213	48.4 [36.1, 56.5]	55.4 [44.3, 64.6]	65.3 [51.6, 77.0]
time after (tercile)	T1	103	2139	43.8 [40.2, 47.8]	52.5 [48.6, 56.8]	60.7 [56.5, 65.1]
time after (tercile)	T2	101	2520	47.1 [43.4, 51.8]	57.0 [54.0, 60.5]	66.2 [63.4, 69.7]
time after (tercile)	T3	102	4919	42.8 [40.3, 45.4]	55.1 [52.1, 58.1]	64.1 [61.1, 66.9]
clip role	ace	17	638	42.3 [36.1, 51.8]	56.4 [50.1, 63.3]	66.0 [58.6, 75.1]
clip role	double_fault	3	13	30.8 [0.0, 100.0]	53.8 [0.0, 100.0]	53.8 [0.0, 100.0]
clip role	first_serve_fault	37	513	45.0 [39.3, 51.6]	56.3 [51.5, 62.1]	64.5 [59.3, 70.1]
clip role	rally	261	8627	44.4 [42.4, 46.6]	54.9 [52.8, 57.0]	63.7 [61.7, 65.9]
score situation	NA	5	148	52.7 [37.5, 62.9]	59.5 [46.5, 71.8]	70.9 [53.6, 87.1]
score situation	break_point	19	684	49.4 [43.9, 55.2]	60.8 [56.2, 66.9]	67.8 [63.8, 73.3]
score situation	deuce_ad	28	1039	42.1 [38.4, 47.1]	53.5 [49.6, 59.0]	63.4 [60.1, 67.3]
score situation	match_point	3	31	67.7 [0.0, 79.2]	54.8 [0.0, 58.3]	64.5 [0.0, 70.8]
score situation	other	257	7632	43.8 [41.7, 45.8]	54.7 [52.6, 56.9]	63.7 [61.5, 66.1]
score situation	set_point	3	172	35.5 [22.5, 61.5]	41.3 [27.0, 68.6]	45.9 [29.7, 76.9]
score situation	tiebreak	3	85	67.1 [44.0, 89.5]	74.1 [56.0, 94.7]	77.6 [64.0, 94.7]

Table 3.6. Per context, 2023 final (held-out match).

context	group	utterances	tokens	in-sample (a) %	pool (a) %	pool (a)+(b) %
phase tag (heuristic)	between_points	13	28	92.9 [76.9, 100.0]	92.9 [76.9, 100.0]	100.0 [100.0, 100.0]
phase tag (heuristic)	changeover	5	110	46.4 [27.6, 72.0]	63.6 [53.8, 81.1]	66.4 [57.5, 83.8]
phase tag (heuristic)	clip	227	7078	43.0 [41.0, 44.9]	59.3 [57.3, 61.1]	67.5 [65.5, 69.3]
dead time before (tercile)	NA	37	777	42.3 [37.6, 48.1]	57.9 [50.3, 64.2]	64.0 [56.3, 70.9]
dead time before (tercile)	T1	70	2251	45.0 [41.2, 48.7]	60.6 [57.7, 63.7]	69.2 [66.3, 71.9]
dead time before (tercile)	T2	69	2014	43.7 [39.7, 47.9]	59.1 [55.6, 62.8]	67.2 [63.3, 71.0]
dead time before (tercile)	T3	69	2174	41.3 [38.1, 44.7]	59.1 [56.0, 62.4]	67.5 [64.4, 70.7]
time after (tercile)	NA	13	271	51.7 [43.4, 64.1]	60.9 [48.1, 78.5]	66.1 [54.1, 82.1]
time after (tercile)	T1	78	1741	40.4 [36.3, 44.7]	57.1 [53.4, 60.7]	65.5 [61.7, 69.3]
time after (tercile)	T2	78	2492	42.2 [39.5, 45.5]	60.1 [57.6, 62.7]	68.5 [66.0, 71.2]
time after (tercile)	T3	76	2712	45.1 [41.7, 48.3]	60.2 [56.9, 63.4]	68.1 [64.6, 71.3]
clip role	ace	5	163	41.7 [31.8, 56.9]	55.8 [49.5, 72.5]	60.1 [54.0, 76.5]
clip role	double_fault	3	99	53.5 [47.6, 65.4]	66.7 [50.0, 83.9]	76.8 [54.8, 96.8]
clip role	first_serve_fault	24	681	44.6 [37.4, 53.4]	56.8 [50.7, 64.0]	65.3 [59.2, 72.9]
clip role	rally	213	6273	42.9 [40.9, 45.0]	59.7 [57.8, 61.7]	67.9 [65.8, 69.8]
score situation	NA	9	161	47.8 [37.9, 59.5]	62.7 [46.6, 78.8]	65.2 [47.3, 83.1]
score situation	break_point	18	686	47.4 [40.3, 54.5]	63.1 [55.8, 69.7]	72.0 [64.3, 79.0]
score situation	deuce_ad	35	1147	46.3 [41.1, 52.1]	59.5 [53.8, 65.1]	68.6 [63.5, 73.5]
score situation	other	178	5102	41.9 [39.7, 44.1]	58.7 [56.6, 60.7]	66.6 [64.6, 68.6]
score situation	set_point	4	106	42.5 [24.4, 75.6]	66.0 [53.4, 92.6]	75.5 [67.4, 95.2]
score situation	tiebreak	1	14	0.0 [0.0, 0.0]	50.0 [50.0, 50.0]	71.4 [71.4, 71.4]

Table 3.7. Sensitivity analyses (2019 unless stated; S-numbers as in plan.md section 8; S5b = revision 1).

variant	design	I -> M	(a) %	(a)+(b) %
S1_text_dedup	D1_in_sample	tv_2019wimF -> tv_2019wimF	43.9 [42.1, 46.0]	48.1 [46.3, 50.2]
S1_text_dedup	D2_split_half	2019 halves -> 2019 other half (pooled)	19.7 [18.4, 21.3]	22.9 [21.4, 24.6]
S1_text_dedup	D3_held_out	pool (18 matches) -> tv_2019wimF	54.6 [52.6, 56.7]	63.6 [61.6, 65.7]
S1_text_dedup	D3_held_out	tv_2019wimF -> tv_2023wimF	25.9 [24.2, 27.6]	29.7 [27.9, 31.6]
S2_no_stop_filter	D1_in_sample	tv_2019wimF -> tv_2019wimF	58.5 [57.0, 59.9]	68.3 [66.9, 69.7]
S2_no_stop_filter	D3_held_out	pool (18 matches) -> tv_2019wimF	70.1 [68.8, 71.5]	83.7 [82.6, 84.7]

variant	design	I -> M	(a) %	(a)+(b) %
S2_no_stop_filter	D3_held_out	tv_2019wimF -> tv_2023wimF	41.8 [40.0, 43.6]	52.2 [50.3, 54.2]
S3_m3	D1_in_sample	tv_2019wimF -> tv_2019wimF	29.6 [28.0, 31.3]	36.5 [34.6, 38.4]
S3_m3	D3_held_out	pool (18 matches) -> tv_2019wimF	48.4 [46.5, 50.4]	59.8 [57.9, 61.9]
S3_m3	D3_held_out	tv_2019wimF -> tv_2023wimF	20.1 [18.7, 21.5]	24.7 [23.1, 26.3]
S4_repeats_ge4_clips_apart	D1_in_sample	tv_2019wimF -> tv_2019wimF	43.9 [42.1, 45.9]	48.1 [46.1, 50.1]
S5_official_calls_masked	D1_in_sample	tv_2019wimF -> tv_2019wimF	43.0 [41.3, 45.0]	47.0 [45.2, 49.0]
S5_official_calls_masked	D2_split_half	2019 halves -> 2019 other half (pooled)	18.9 [17.5, 20.4]	21.9 [20.2, 23.5]
S5_official_calls_masked	D3_held_out	pool (18 matches) -> tv_2019wimF	54.4 [52.4, 56.3]	63.1 [61.0, 65.0]
S5_official_calls_masked	D3_held_out	tv_2019wimF -> tv_2023wimF	25.0 [23.5, 26.8]	28.5 [26.8, 30.2]
S6_nmin3	D1_in_sample	tv_2019wimF -> tv_2019wimF	19.4 [18.0, 20.9]	29.8 [28.0, 31.8]
S6_nmin3	D3_held_out	pool (18 matches) -> tv_2019wimF	24.4 [22.7, 26.1]	47.0 [44.9, 49.1]
S6_nmin3	D3_held_out	tv_2019wimF -> tv_2023wimF	6.9 [5.8, 8.0]	14.4 [12.9, 16.0]
S6_nmin3	D2_split_half	2019 halves -> 2019 other half (pooled)	4.5 [3.7, 5.3]	11.1 [9.8, 12.4]
S7_blocks_of_20_clips	D2_split_half	2019 halves -> 2019 other half (pooled)	19.8 [18.3, 21.2]	22.4 [20.9, 24.1]
S5b_umpire_and_score_calls_masked	D1_in_sample	tv_2019wimF -> tv_2019wimF	41.5 [39.7, 43.3]	45.3 [43.4, 47.2]
S5b_umpire_and_score_calls_masked	D2_split_half	2019 halves -> 2019 other half (pooled)	17.8 [16.4, 19.2]	19.8 [18.4, 21.3]
S5b_umpire_and_score_calls_masked	D3_held_out	pool (18 matches) -> tv_2019wimF	53.3 [51.4, 55.2]	61.8 [59.7, 63.7]
S5b_umpire_and_score_calls_masked	D3_held_out	tv_2019wimF -> tv_2023wimF	24.0 [22.4, 25.7]	27.2 [25.5, 29.0]
D7_shuffled_word_baseline_R200	D1_in_sample	2019 shuffled words -> 2019 shuffled words	16.2 [15.2, 17.3]	22.2 [21.0, 23.4]
D7_shuffled_word_baseline_R200	D2_split_half	2019 shuffled words -> 2019 shuffled words	2.8 [2.2, 3.5]	4.4 [3.6, 5.2]
S8_random_half_splits_R200	D2_split_half	2019 random halves -> 2019 other half (pooled)	20.6 [19.5, 21.6]	23.4 [22.3, 24.4]

S5 masks 2.0% of 2019 tokens as umpire/Hawk-Eye/announcer calls (score calls not masked); S5b, the revised mask that adds score calls, `thank you` and the other umpire forms (section 1), masks 4.6%.

Table 3.8. POST HOC exploratory: injected ASR-like substitution noise (added after C1/C2 were seen; plan.md addenda 1-2). Each token replaced with probability e by

a token drawn from the TV-pool unigram distribution, in both I and M; (a) coverage, mean [2.5-97.5%] over 50 (split-half) or 50 (held-out) replicates. e = 0.022 is the corpus's hand-read lower bound on the TV error rate. For TV rows e is noise added on top of the ASR errors already present; in the TV held-out row I and M are fixed (one pool, one final), so its range reflects only the noise draws, whereas the Cornell and press rows also resample I and M.

design, corpus	e = 0.0	e = 0.022	e = 0.05	e = 0.1	e = 0.15
split-half, TV pool subsample	16.3 [14.9, 17.9]	15.3 [13.7, 16.6]	13.9 [12.5, 15.5]	12.0 [10.7, 13.6]	10.1 [9.0, 11.4]
split-half, Cornell text	35.2 [33.1, 37.3]	33.7 [31.6, 36.2]	31.3 [28.7, 33.7]	26.9 [25.0, 28.7]	23.1 [20.7, 25.6]
split-half, press answers	24.7 [22.5, 26.5]	23.0 [20.6, 24.8]	21.4 [19.2, 23.3]	18.6 [17.0, 20.4]	15.9 [14.1, 17.7]
held-out, TV pool -> 2019	-	53.1 [52.8, 53.5]	50.8 [50.1, 51.2]	47.2 [46.4, 47.9]	43.9 [43.1, 44.9]
held-out, Cornell text (I median 65,444, Table 3.2)	-	68.9 [66.4, 71.5]	65.5 [63.1, 67.8]	60.6 [58.6, 62.9]	55.6 [53.6, 57.9]
held-out, press answers	-	63.7 [61.7, 65.6]	61.4 [59.4, 63.3]	56.9 [54.7, 58.8]	52.6 [50.3, 54.1]

Crossing points (linear interpolation between grid levels; TV references: split-half TV pool at e = 0, and pool -> 2019): split-half, press: mean reaches the TV value (16.3%) at e* = 0.142; ranges first overlap at e = 0.1; held-out, press: mean reaches the TV value (55.0%) at e* = 0.122; ranges first overlap at e = 0.1; split-half, Cornell: mean stays above the TV value (16.3%) up to e = 0.15; ranges do not overlap up to e = 0.15; held-out, Cornell: mean stays above the TV value (55.0%) up to e = 0.15; ranges first overlap at e = 0.15. The noise model is not a bound on the real effect of ASR (section 9.1).

4. Confirmatory tests and replication

Revised reporting (plan.md addendum 2): C1/R1 are indeterminate whatever their statistics (A1); for C1/R1/C2 the inferential statistic is the 95% interval of the replicate differences, and the pre-registered "p" (2 x the share of replicate pairs on the minority side) is shown only with its floor 2/(pairs + 1), because it is not a calibrated null test (B1); C4/C5/R4/R5 use the commentary-only references (B6; the pre-registered all-token results are in Table 4.1b). Holm-Bonferroni is computed, as pre-registered, within the 2019 family (C1, C2, C3a, C3b, C4, C5) and within the 2023 replication family; it is informative only for the permutation tests C3-C5.

test	hypothesis / statistic	estimate	95% interval	p (pre-registered statistic)	p (Holm)	verdict
C1	(a) repeated-n-gram coverage, pool -> 2019 final, minus press answers (I ~100k -> M 9.7k, disjoint interviewees)	55.0 vs 66.1; diff -11.1 pp	[-13.9, -8.0] pp	0.0002 (= floor 0.0002)	0.0012	indeterminate pending a WER estimate
R1	(a) repeated-n-gram coverage, pool -> 2023 final, minus press answers (I ~100k -> M 9.7k, disjoint interviewees)	59.5 vs 66.1; diff -6.7 pp	[-9.5, -3.6] pp	0.0002 (= floor 0.0002)	0.0010	indeterminate pending a WER estimate
C2	corpus contrast, split-half (a) repeated-n-gram coverage at 2019 size: TV pool subsamples minus Cornell live-text subsamples	16.2 vs 35.3; diff -19.1 pp	[-22.1, -16.3] pp	0.0020 (= floor 0.0020)	0.0100	corpus difference: 95% interval of differences excludes 0 (not attributable to medium)
C3a	(a) coverage (pool-identified), shortest minus longest tercile of dead_time_before	55.4 vs 55.1; diff 0.4 pp	[-4.0, 4.6] pp	0.8770	1.0000	not rejected; MDE (80% power, alpha 0.05) = 6.4 pp
C3b	(a) coverage (pool-identified), shortest minus longest tercile of time_after	52.5 vs 55.1; diff -2.5 pp	[-7.3, 2.6] pp	0.3029	0.9088	not rejected; MDE (80% power, alpha 0.05) = 7.0 pp
R3a	(a) coverage (pool-identified), shortest minus longest tercile of dead_time_before	60.6 vs 59.1; diff 1.5 pp	[-2.9, 5.9] pp	0.5199	0.6947	not rejected; MDE (80% power, alpha 0.05) = 6.5 pp

test	hypothesis / statistic	estimate	95% interval	p (pre-registered statistic)	p (Holm)	verdict
R3b	(a) coverage (pool-identified), shortest minus longest tercile of time_after	57.1 vs 60.2; diff -3.1 pp	[-8.2, 1.9] pp	0.2304	0.6947	not rejected; MDE (80% power, alpha 0.05) = 7.2 pp
C4	thrift: distinct expressions per player x slot x dead-time tercile below permutation null (commentary_only)	D = 86 vs null mean 84.2	null: 81-88	0.8932	1.0000	not rejected; MDE (80% power, alpha 0.05): theta = 0.30
R4	thrift: distinct expressions per player x slot x dead-time tercile below permutation null (commentary_only)	D = 47 vs null mean 49.0	null: 46-52	0.1737	0.6947	not rejected; MDE (80% power, alpha 0.05): theta = 0.42
C5	extension: Spearman rho of expression length (syllables) with dead_time_before_s (commentary_only)	rho = 0.090	null: -0.117 to 0.114	0.1295	0.5179	not rejected; MDE (80% power, alpha 0.05): rho = 0.17
R5	extension: Spearman rho of expression length (syllables) with dead_time_before_s (commentary_only)	rho = 0.076	null: -0.165 to 0.147	0.3453	0.6947	not rejected; MDE (80% power, alpha 0.05): rho = 0.22

- **C1/R1** (commentary more repetitive than non-commentary tennis speech, held-out): indeterminate pending a WER estimate. The difference interval ([-13.9, -8.0] pp for 2019, [-9.5, -3.6] pp for 2023) excludes 0, but the comparison confounds commentary with ASR noise, transcription convention and held-out design (section 9.1).
- **C2** (corpus contrast, split-half at matched size, TV pool vs Cornell live text): corpus difference: 95% interval of differences excludes 0 (not attributable to medium); TV pool 16.2% vs Cornell 35.3%, difference -19.1 pp [-22.1, -16.3] over 1000 independent subsample pairs (R raised from 200 in v1 to 1000; f04 stage runtime 6.3 min on 4 CPUs). Its p is the replicate-overlap share at its floor 2/(1000 + 1) = 0.0020: every one of the 1000 paired differences has the same sign. In v1, with R = 200, the same statistic was 0.00995 and its Holm value 0.0498 (rank 2 of 6); that value reflected R, not evidence near the 0.05 boundary.
- **C3a/C3b** (less time, more repetition): not rejected / not rejected (MDE 6.4 / 7.0 pp, Table 4.3). Shortest minus longest tercile: dead time before 0.4 pp [-4.0, 4.6]; time after -2.5 pp [-7.3, 2.6]. C3a's predictor (`dead_time_before_s`) is the interval before the point, when the clip's text has not started; C3b (`time_after_s`) is the apt one (section 9.2).
- **C4** (thrift bound to available time, commentary-only references): not rejected; D = 86, null mean 84.2 (MDE theta = 0.30).
- **C5** (longer expressions with more time, commentary-only references): not rejected; rho = 0.090 (MDE rho = 0.17).
- Replication on the 2023 final: R1 indeterminate pending a WER estimate, R3a not rejected, R3b not rejected, R4 not rejected, R5 not rejected.

Table 4.1b. Sensitivity: C4/C5/R4/R5 on the pre-registered token set (all references, including umpire-pattern ones; these are the v1 confirmatory results; Holm in the original family composition).

test	hypothesis / statistic	estimate	null 95%	p	p (Holm)	verdict
C4_all_tokens	thrift: distinct expressions per player x slot x dead-time tercile below permutation null (all_tokens)	D = 91 vs null mean 89.6	null: 86-93	0.8403	1.0000	not rejected
R4_all_tokens	thrift: distinct expressions per player x slot x dead-time tercile below permutation null (all_tokens)	D = 59 vs null mean 60.3	null: 57-63	0.3052	0.9214	not rejected
C5_all_tokens	extension: Spearman rho of expression length (syllables) with dead_time_before_s (all_tokens)	rho = 0.087	null: -0.108 to 0.111	0.1173	0.4692	not rejected
R5_all_tokens	extension: Spearman rho of expression length (syllables) with dead_time_before_s (all_tokens)	rho = 0.024	null: -0.157 to 0.127	0.7564	1.0000	not rejected

Table 4.3. Minimum detectable effects (post hoc, `f09_power.py`; 400 simulated data sets per grid point; 80% power; simulation models in the script's docstring). 'Type I at 0' = rejection rate with no injected effect (should be near 0.05).

test	effect scale	observed	MDE, alpha = 0.05	MDE, Bonferroni alpha	type I at 0
C3a	coverage difference T1 - T3 (pp)	0.4 [-4.0, 4.6]	6.4	7.9 (alpha = 0.0083)	0.04
C3b	coverage difference T1 - T3 (pp)	-2.5 [-7.3, 2.6]	7.0	8.5 (alpha = 0.0083)	0.06
R3a	coverage difference T1 - T3 (pp)	1.5 [-2.9, 5.9]	6.5	7.9 (alpha = 0.0100)	0.03
R3b	coverage difference T1 - T3 (pp)	-3.1 [-8.2, 1.9]	7.2	8.5 (alpha = 0.0100)	0.05
C4	theta: share of references with a tercile-specific form (Cramer's V)	V = 0.31 (null V = 0.30)	theta = 0.30 (V = 0.45; D 5.8 below its null mean)	theta = 0.38 (V = 0.51) (alpha = 0.0083)	0.04
R4	theta: share of references with a tercile-specific form (Cramer's V)	V = 0.32 (null V = 0.31)	theta = 0.42 (V = 0.52; D 5.0 below its null mean)	theta = 0.48 (V = 0.57) (alpha = 0.0100)	0.03
C5	Spearman rho, syllables vs dead time	rho = 0.090	rho = 0.17 (latent r = 0.20)	rho = 0.21 (alpha = 0.0083)	0.04
R5	Spearman rho, syllables vs dead time	rho = 0.076	rho = 0.22 (latent r = 0.30)	rho = 0.27 (alpha = 0.0100)	0.05

Simulation runtime 1.0 min. The C3 MDEs are about 6-7 pp on a base of about 55%; C4 needs roughly 30% of references to take a tercile-specific form (in a design where the surname is the commonest form in every stratum); C5 needs rho of about 0.17 with ties as observed.

Table 4.4. Serial correlation check (post hoc, exploratory): circular-shift nulls (the context sequence shifted against the text sequence by at least 10 utterances, all shifts) beside the label permutations used in the tests.

test	observed	p, label permutation	p, circular shift	shifts	null SD, permutation / shift
C3a	0.4 pp	0.8828	0.9033	299	2.33 / 2.63 pp
C3b	-2.5 pp	0.3149	0.3533	299	2.51 / 2.60 pp
R3a	1.5 pp	0.5274	0.5815	226	2.33 / 2.57 pp
R3b	-3.1 pp	0.2298	0.2070	226	2.60 / 2.53 pp
C5	rho = 0.090	0.1295	0.2267	299	-
R5	rho = 0.076	0.3453	0.3436	226	-

Table 4.2. Exploratory corpus contrasts (difference x - y in percentage points, 95% interval over replicate pairs; the p column is the same replicate-overlap share as for C1/C2, with its floor, and is not a calibrated test).

id	contrast	x %	y %	diff [95%]	p (floor)
E-C1-text	(a) coverage, pool -> 2019 final, minus Cornell text held-out (disjoint player pairs; I median 65,444 tokens, not 100k)	55.0	70.9	-15.8 [-19.2, -12.2]	0.0002 (0.0002)
E-C1-press-groupedM	(a) coverage, pool -> 2019 final, minus press held-out with M concentrated in few groups, I ~100k (D5ii-b, post hoc)	55.0	66.4	-11.3 [-18.1, -5.4]	0.0002 (0.0002)
E-C1-text-groupedM	(a) coverage, pool -> 2019 final, minus text held-out with M concentrated in few groups, I ~100k (D5ii-b, post hoc)	55.0	76.0	-21.0 [-25.3, -15.9]	0.0002 (0.0002)
E-R1-text	(a) coverage, pool -> 2023 final, minus Cornell text held-out (disjoint player pairs; I median 65,444 tokens, not 100k)	59.5	70.9	-11.4 [-14.8, -7.8]	0.0002 (0.0002)
E-R1-press-groupedM	(a) coverage, pool -> 2023 final, minus press held-out with M concentrated in few groups, I ~100k (D5ii-b, post hoc)	59.5	66.4	-6.9 [-13.6, -0.9]	0.0090 (0.0002)
E-R1-text-groupedM	(a) coverage, pool -> 2023 final, minus text held-out with M concentrated in few groups, I ~100k (D5ii-b, post hoc)	59.5	76.0	-16.5 [-20.8, -11.6]	0.0002 (0.0002)

id	contrast	x %	y %	diff [95%]	p (floor)
E-split-tv_pool-vs-press	split-half (a) coverage at 2019 size: tv_pool minus press	16.2	24.6	-8.4 [-11.1, -5.8]	0.0020 (0.0020)
E-split-cornell_text-vs-press	split-half (a) coverage at 2019 size: cornell_text minus press	35.3	24.6	10.7 [7.1, 14.0]	0.0020 (0.0020)
E-insample_a-tv-vs-text	insample_a at 2019 size: TV pool minus Cornell text	38.9	60.7	-21.8 [-24.8, -18.9]	0.0020 (0.0020)
E-insample_a-tv-vs-press	insample_a at 2019 size: TV pool minus press answers	38.9	48.6	-9.7 [-12.8, -6.5]	0.0020 (0.0020)
E-insample_ab-tv-vs-text	insample_ab at 2019 size: TV pool minus Cornell text	42.4	66.5	-24.2 [-27.0, -21.1]	0.0100 (0.0100)
E-insample_ab-tv-vs-press	insample_ab at 2019 size: TV pool minus press answers	42.4	52.8	-10.4 [-13.8, -7.2]	0.0100 (0.0100)
E-splithalf_ab-tv-vs-text	splithalf_ab at 2019 size: TV pool minus Cornell text	17.7	41.0	-23.3 [-26.1, -20.4]	0.0100 (0.0100)
E-splithalf_ab-tv-vs-press	splithalf_ab at 2019 size: TV pool minus press answers	17.7	26.3	-8.6 [-11.3, -5.8]	0.0100 (0.0100)
E-insample_n3-tv-vs-text	insample_n3 at 2019 size: TV pool minus Cornell text	15.6	35.8	-20.2 [-23.2, -16.9]	0.0020 (0.0020)
E-insample_n3-tv-vs-press	insample_n3 at 2019 size: TV pool minus press answers	15.6	25.3	-9.7 [-12.9, -6.5]	0.0020 (0.0020)
E-splithalf_n3-tv-vs-text	splithalf_n3 at 2019 size: TV pool minus Cornell text	3.6	14.2	-10.6 [-13.0, -8.2]	0.0020 (0.0020)
E-splithalf_n3-tv-vs-press	splithalf_n3 at 2019 size: TV pool minus press answers	3.6	8.2	-4.6 [-6.6, -2.7]	0.0020 (0.0020)
E-splithalf_n4-tv-vs-text	splithalf_n4 at 2019 size: TV pool minus Cornell text	0.4	3.5	-3.1 [-4.5, -1.9]	0.0020 (0.0020)
E-splithalf_n4-tv-vs-press	splithalf_n4 at 2019 size: TV pool minus press answers	0.4	1.0	-0.6 [-1.5, 0.1]	0.1219 (0.0020)
E-splithalf_content-tv-vs-text	splithalf_content at 2019 size: TV pool minus Cornell text	15.1	31.9	-16.8 [-19.6, -13.9]	0.0020 (0.0020)
E-splithalf_content-tv-vs-press	splithalf_content at 2019 size: TV pool minus press answers	15.1	24.0	-8.9 [-11.5, -6.2]	0.0020 (0.0020)
E-splithalf_content_n3-tv-vs-text	splithalf_content_n3 at 2019 size: TV pool minus Cornell text	3.4	13.7	-10.3 [-12.6, -8.0]	0.0020 (0.0020)
E-splithalf_content_n3-tv-vs-press	splithalf_content_n3 at 2019 size: TV pool minus press answers	3.4	8.0	-4.6 [-6.5, -2.7]	0.0020 (0.0020)
E-insample_content_n3-tv-vs-text	insample_content_n3 at 2019 size: TV pool minus Cornell text	15.0	34.2	-19.1 [-22.1, -15.9]	0.0020 (0.0020)
E-insample_content_n3-tv-vs-press	insample_content_n3 at 2019 size: TV pool minus press answers	15.0	24.9	-9.9 [-13.1, -6.7]	0.0020 (0.0020)

5. Noun-epithet systems

Referring expressions found: 2019 Federer 164, Djokovic 188; 2023 Djokovic 121, Alcaraz 80. Third-person masculine pronouns (excluded from the systems, counted): 2019 288, 2023 247. 25 descriptive-epithet occurrences were assigned to a referent by hand verdict (`hand/epithet_referents.tsv`); 0 unresolved.

Umpire-pattern references (revision 1, rules in section 1), excluded from the thrift and extension tests: 2019 `advantage_name` 2; 2019 `challenging` 1; 2019 `game_name` 9; 2019

`leads_games` 8; 2019 `mr_surname` 11; 2023 `advantage_name` 7; 2023 `challenges_remaining` 7; 2023 `challenging` 9; 2023 `leads_games` 8.

Inventory: full name, `mr` + surname (umpire), surname, first name, hypocoristic (`nole`, `rog`, `carlitos`), and descriptive epithets matched by a fixed regex family (`refexpr.py`). Slot: spaCy `en_core_web_sm 3.8.0` dependency label of the expression's head, mapped as in plan.md section 5. Syllables: CMU Pronouncing Dictionary (vowel phonemes), vowel-group fallback for names not in it (`djokovic` 3; `nole` gets 1, which is wrong: it has two), digits spelled out.

Slot accuracy against my hand labels on a random sample: 33/40 = 82.5% (Wilson 95% 68.0-91.3%). Disagreements (hand|parser): object|subject 1, other|after_preposition 1, other|object 1, subject|after_preposition 1, subject|object 1, subject|vocative_exclamation 1, vocative_exclamation|object 1.

Audit: strings of the form `the (word){0,3} <person noun>` that the inventory does not match (count >= 2, or containing champion / number one / seed / player / returner), from `results/epithet_discovery.csv`; they are not added, to keep the pre-specified inventory: 2019 `the man` 6; 2019 `the great` 4; 2019 `the player` 3; 2019 `the best returner` 2; 2019 `the greatest` 2; 2019 `the last one` 2; 2019 `the one` 2; 2019 `the better player` 1; 2019 `the highest earning player` 1; 2019 `the match player` 1; 2019 `the number 21 seed` 1; 2019 `the number one` 1; 2019 `the size of the player` 1; 2019 `the year end number one` 1; 2023 `the greatest` 3; 2023 `the first one` 2; 2023 `the best difficult weather player` 1; 2023 `the four time defending champion` 1; 2023 `the player` 1.

Table 5.1. Every referring expression by player, with syllables and syntactic slot (all references; 'umpire' = of which inside an umpire pattern).

match	player	category	expression	tokens	umpire	syll.	subj	obj	poss	after prep	voc/excl	other
2019	djokovic	surname	djokovic	127	11	3	50	10	6	42	10	9
2019	djokovic	first_name	novak	29	0	2	14	3	0	7	3	2
2019	djokovic	full_name	novak djokovic	17	0	5	5	4	1	5	1	1
2019	djokovic	title_surname	mr djokovic	5	5	5	5	0	0	0	0	0
2019	djokovic	epithet	the world number one	3	0	5	0	1	1	1	0	0
2019	djokovic	epithet	the top seed	1	0	3	0	0	0	0	0	1
2019	djokovic	epithet	the young man	1	0	3	0	0	0	1	0	0
2019	djokovic	epithet	the number one seed	1	0	5	0	1	0	0	0	0
2019	djokovic	epithet	the defending champion	1	0	7	0	1	0	0	0	0
2019	djokovic	epithet	the serbian	1	0	4	0	0	0	1	0	0
2019	djokovic	hypocoristic	nole	1	0	1	1	0	0	0	0	0
2019	djokovic	epithet	the younger man	1	0	4	0	1	0	0	0	0
2019	federer	surname	federer	127	9	3	54	13	12	24	8	16
2019	federer	first_name	roger	17	0	2	10	0	1	5	1	0
2019	federer	full_name	roger federer	8	0	5	3	2	0	3	0	0
2019	federer	title_surname	mr federer	6	6	5	6	0	0	0	0	0
2019	federer	epithet	the great swiss	3	0	3	1	0	0	2	0	0
2019	federer	epithet	the man from basel	1	0	5	0	0	0	1	0	0

match	player	category	expression	tokens	umpire	syll.	subj	obj	poss	after prep	voc/excl	other
2019	federer	epithet	the swiss	1	0	2	0	0	0	1	0	0
2019	federer	epithet	the older man	1	0	4	1	0	0	0	0	0
2023	alcaraz	surname	alcaraz	61	9	3	28	4	0	19	5	5
2023	alcaraz	full_name	carlos alcaraz	10	10	5	8	1	0	0	0	1
2023	alcaraz	first_name	carlos	5	0	2	4	0	0	1	0	0
2023	alcaraz	epithet	the world number one	2	0	5	0	1	0	1	0	0
2023	alcaraz	epithet	the young spaniard	1	0	4	0	1	0	0	0	0
2023	alcaraz	epithet	the young man	1	0	3	0	0	0	1	0	0
2023	djokovic	surname	djokovic	86	6	3	29	8	2	29	12	6
2023	djokovic	first_name	novak	18	0	2	8	4	1	5	0	0
2023	djokovic	full_name	novak djokovic	12	6	5	5	0	0	2	1	4
2023	djokovic	epithet	the champion	4	0	4	1	1	0	2	0	0
2023	djokovic	epithet	the serb	1	0	2	0	0	0	1	0	0

Table 5.2. Length in syllables (all references).

match	player	syllables	tokens	expressions
2019	djokovic	1	1	nole:1
2019	djokovic	2	29	novak:29
2019	djokovic	3	129	djokovic:127; the top seed:1; the young man:1
2019	djokovic	4	2	the serbian:1; the younger man:1
2019	djokovic	5	26	novak djokovic:17; mr djokovic:5; the world number one:3; the number one seed:1
2019	djokovic	7	1	the defending champion:1
2019	federer	2	18	roger:17; the swiss:1
2019	federer	3	130	federer:127; the great swiss:3
2019	federer	4	1	the older man:1
2019	federer	5	15	roger federer:8; mr federer:6; the man from basel:1
2023	alcaraz	2	5	carlos:5
2023	alcaraz	3	62	alcaraz:61; the young man:1
2023	alcaraz	4	1	the young spaniard:1
2023	alcaraz	5	12	carlos alcaraz:10; the world number one:2
2023	djokovic	2	19	novak:18; the serb:1
2023	djokovic	3	86	djokovic:86
2023	djokovic	4	4	the champion:4
2023	djokovic	5	12	novak djokovic:12

Table 5.3. Timing context (all references; dead-time and time-after terciles, phase and score situation; tokens, distinct expressions, mean syllables, share of bare surname).

match	player	context	group	tokens	distinct	mean syll.	surname %
2019	djokovic	phase	between_points	2	1	3.000	100
2019	djokovic	phase	changeover	2	2	2.500	50
2019	djokovic	phase	clip	184	12	3.152	67
2019	federer	phase	changeover	3	1	3.000	100
2019	federer	phase	clip	161	8	3.081	77
2023	alcaraz	phase	between_points	1	1	3.000	100
2023	alcaraz	phase	changeover	3	1	3.000	100
2023	alcaraz	phase	clip	76	6	3.263	75
2023	djokovic	phase	between_points	2	1	3.000	100
2023	djokovic	phase	changeover	4	2	3.250	75
2023	djokovic	phase	clip	115	5	3.070	70

match	player	context	group	tokens	distinct	mean syll.	surname %
2019	djokovic	dtb_terc	NA	17	4	3.294	59
2019	djokovic	dtb_terc	T1	53	6	2.962	62
2019	djokovic	dtb_terc	T2	56	5	3.125	77
2019	djokovic	dtb_terc	T3	62	9	3.274	66
2019	federer	dtb_terc	NA	17	4	3.235	82
2019	federer	dtb_terc	T1	52	7	3.077	75
2019	federer	dtb_terc	T2	48	5	3.021	81
2019	federer	dtb_terc	T3	47	5	3.085	74
2023	alcaraz	dtb_terc	NA	7	2	3.286	86
2023	alcaraz	dtb_terc	T1	29	4	3.172	76
2023	alcaraz	dtb_terc	T2	26	5	3.423	69
2023	alcaraz	dtb_terc	T3	18	4	3.111	83
2023	djokovic	dtb_terc	NA	10	4	3.100	60
2023	djokovic	dtb_terc	T1	38	4	2.895	76
2023	djokovic	dtb_terc	T2	27	3	3.296	74
2023	djokovic	dtb_terc	T3	46	5	3.087	67
2019	djokovic	ta_terc	NA	5	4	3.400	20
2019	djokovic	ta_terc	T1	35	6	3.029	54
2019	djokovic	ta_terc	T2	60	8	3.083	72
2019	djokovic	ta_terc	T3	88	6	3.216	73
2019	federer	ta_terc	NA	5	1	3.000	100
2019	federer	ta_terc	T1	27	5	2.963	59
2019	federer	ta_terc	T2	49	5	3.122	80
2019	federer	ta_terc	T3	83	7	3.096	81
2023	alcaraz	ta_terc	NA	2	1	3.000	100
2023	alcaraz	ta_terc	T1	19	4	3.526	58
2023	alcaraz	ta_terc	T2	27	3	3.111	89
2023	alcaraz	ta_terc	T3	32	6	3.219	75
2023	djokovic	ta_terc	NA	6	3	3.500	67
2023	djokovic	ta_terc	T1	27	4	2.852	63
2023	djokovic	ta_terc	T2	39	4	3.051	72
2023	djokovic	ta_terc	T3	49	4	3.163	76
2019	djokovic	score	NA	3	3	3.333	33
2019	djokovic	score	break_point	17	5	3.118	65
2019	djokovic	score	deuce_ad	23	4	3.217	70
2019	djokovic	score	match_point	2	1	3.000	100
2019	djokovic	score	other	137	11	3.139	66
2019	djokovic	score	set_point	3	1	3.000	100
2019	djokovic	score	tiebreak	3	1	3.000	100
2019	federer	score	NA	4	1	3.000	100
2019	federer	score	break_point	11	2	3.182	91
2019	federer	score	deuce_ad	19	4	3.158	79
2019	federer	score	match_point	2	1	3.000	100
2019	federer	score	other	125	8	3.064	74
2019	federer	score	set_point	2	1	3.000	100
2019	federer	score	tiebreak	1	1	3.000	100
2023	alcaraz	score	NA	2	1	3.000	100
2023	alcaraz	score	break_point	12	4	3.500	67
2023	alcaraz	score	deuce_ad	11	1	3.000	100
2023	alcaraz	score	other	53	5	3.245	76
2023	alcaraz	score	set_point	2	2	3.500	0
2023	djokovic	score	NA	4	2	3.250	75
2023	djokovic	score	break_point	12	3	3.417	67
2023	djokovic	score	deuce_ad	20	3	3.200	75
2023	djokovic	score	other	85	5	2.988	71

Table 5.4. Pronouns (he, him, his, himself, he's, he'd, he'll) by dead-time tercile; attribution is heuristic (the single player named in the same utterance).

match	group	attributed to	utterances	pronouns
2019	all	ANY	318	288
2019	all	djokovic	63	55
2019	all	federer	37	31
2019	all	unattributed	218	202
2023	all	ANY	245	247
2023	all	alcaraz	29	28
2023	all	djokovic	64	73
2023	all	unattributed	152	146
2019	NA	ANY	36	26
2019	NA	djokovic	9	13
2019	NA	federer	4	2
2019	NA	unattributed	23	11
2019	T1	ANY	94	75
2019	T1	djokovic	16	13
2019	T1	federer	12	8
2019	T1	unattributed	66	54
2019	T2	ANY	94	86
2019	T2	djokovic	16	13
2019	T2	federer	10	6
2019	T2	unattributed	68	67
2019	T3	ANY	94	101
2019	T3	djokovic	22	16
2019	T3	federer	11	15
2019	T3	unattributed	61	70
2023	NA	ANY	37	28
2023	NA	alcaraz	5	6
2023	NA	djokovic	7	14
2023	NA	unattributed	25	8
2023	T1	ANY	70	75
2023	T1	alcaraz	13	6
2023	T1	djokovic	20	25
2023	T1	unattributed	37	44
2023	T2	ANY	69	76
2023	T2	alcaraz	9	11
2023	T2	djokovic	13	9
2023	T2	unattributed	47	56
2023	T3	ANY	69	68
2023	T3	alcaraz	2	5
2023	T3	djokovic	24	25
2023	T3	unattributed	43	38

6. Economy (thrift) and extension

Functionally equivalent = different expression types (normalised wording, possessive stripped) referring to the same player in the same syntactic slot and the same context cell. D = sum over player x slot x context cells of the number of distinct types. Null: context labels permuted among tokens within each player x slot stratum (10,000 permutations); one-sided p for 'fewer types than chance' (thrift). The utterance-level permutation (exploratory) keeps tokens of one utterance together. Revision 1: the tests are run on the commentary-only references (Table 6.1); the pre-registered all-reference results are in Table 6.1b and Table 4.1b.

Table 6.1. Thrift tests, commentary-only references.

match	context	players	permutation	status	tokens	cells	D obs	null mean [95%]	p (fewer)	p (two-sided)	single-type cells %
2019	dtb_terc	both	token_within_player_slot	confirmatory	290	36	86	84.2 [81, 88]	0.8932	0.4167	25
2019	dtb_terc	both	utterance	exploratory	290	36	86	82.8 [79, 87]	0.9668	0.1516	25
2019	dtb_terc	djokovic	token_within_player_slot	exploratory	156	18	50	48.3 [45, 51]	0.9364	0.3167	6
2019	dtb_terc	federer	token_within_player_slot	exploratory	134	18	36	35.9 [34, 38]	0.6930	1.0000	44
2019	ta_terc	both	token_within_player_slot	exploratory	312	35	80	83.2 [79, 87]	0.0811	0.1244	37
2019	phase	both	token_within_player_slot	exploratory	321	15	49	48.9 [48, 50]	0.8151	1.0000	20
2019	score	both	token_within_player_slot	exploratory	314	41	81	81.7 [77, 86]	0.4678	0.8147	54
2019	role	both	token_within_player_slot	exploratory	321	27	61	62.8 [60, 66]	0.2118	0.3673	48
2023	dtb_terc	both	token_within_player_slot	confirmatory	154	27	47	49.0 [46, 52]	0.1737	0.2291	48
2023	dtb_terc	both	utterance	exploratory	154	27	47	50.8 [47, 54]	0.0380	0.0525	48
2023	dtb_terc	alcaraz	token_within_player_slot	exploratory	55	12	19	19.0 [18, 20]	0.7289	1.0000	42
2023	dtb_terc	djokovic	token_within_player_slot	exploratory	99	15	28	30.0 [27, 33]	0.1513	0.2853	53
2023	ta_terc	both	token_within_player_slot	exploratory	163	27	51	51.2 [48, 54]	0.5794	1.0000	41
2023	phase	both	token_within_player_slot	exploratory	170	17	34	33.4 [32, 34]	1.0000	0.6002	53
2023	score	both	token_within_player_slot	exploratory	164	27	46	45.9 [43, 49]	0.6561	1.0000	56
2023	role	both	token_within_player_slot	exploratory	170	18	37	36.5 [34, 39]	0.7907	1.0000	44

Of the 28 exploratory thrift rows (both reference sets), 1 has an unadjusted one-sided p below 0.05: commentary_only 2023 dtb_terc both utterance p = 0.0380. With this many exploratory rows such values are expected by chance and are not interpreted.

Table 6.1b. Sensitivity: thrift tests on all references (pre-registered token set; exploratory rows in `results/thrift_tests.csv`).

match	context	players	permutation	status	tokens	cells	D obs	null mean [95%]	p (fewer)	p (two-sided)	single-type cells %
2019	dtb_terc	both	token_within_player_slot	sensitivity (pre-registered token set)	318	36	91	89.6 [86, 93]	0.8403	0.6003	25
2023	dtb_terc	both	token_within_player_slot	sensitivity (pre-	184	31	59	60.3 [57,	0.3052	0.5415	39

match	context	players	permutation	status	tokens	cells	D obs	null mean [95%]	p (fewer)	p (two-sided)	single-type cells %
				registered token set)				63]			

Table 6.2. Extension (size and range of each player's system).

references	match	player	tokens	distinct expressions	categories	syllables min-max	distinct lengths	occupied cells (slot x dead-time x length class)	surname %	epit %
commentary_only	2019	djokovic	172	11	5	1-7	6	50	67	5
commentary_only	2019	federer	149	7	4	2-5	4	38	79	4
all_tokens	2019	djokovic	188	12	6	1-7	6	52	68	5
all_tokens	2019	federer	164	8	5	2-5	4	39	77	4
commentary_only	2023	alcaraz	61	5	3	2-5	4	21	85	7
commentary_only	2023	djokovic	109	5	4	2-5	4	35	73	5
all_tokens	2023	alcaraz	80	6	4	2-5	4	30	76	5
all_tokens	2023	djokovic	121	5	4	2-5	4	39	71	4

Table 6.3. Length vs available time (Spearman rho over expression tokens).

references	match	time	permutation	status	tokens	rho	null 95%	p
commentary_only	2019	dead_time_before_s	token_within_player	confirmatory	290	0.090	-0.117 to 0.114	0.1295
commentary_only	2019	dead_time_before_s	utterance	exploratory	290	0.090	-0.138 to 0.134	0.2015
commentary_only	2019	time_after_s	token_within_player	exploratory	312	0.103	-0.113 to 0.106	0.0672
commentary_only	2019	time_after_s	utterance	exploratory	312	0.103	-0.128 to 0.128	0.1211
all_tokens	2019	dead_time_before_s	token_within_player	sensitivity (pre-registered token set)	318	0.087	-0.108 to 0.111	0.1173
all_tokens	2019	dead_time_before_s	utterance	exploratory	318	0.087	-0.127 to 0.129	0.1897
all_tokens	2019	time_after_s	token_within_player	exploratory	342	0.085	-0.110 to 0.104	0.1219
all_tokens	2019	time_after_s	utterance	exploratory	342	0.085	-0.122 to 0.120	0.1817
commentary_only	2023	dead_time_before_s	token_within_player	confirmatory	154	0.076	-0.165 to 0.147	0.3453
commentary_only	2023	dead_time_before_s	utterance	exploratory	154	0.076	-0.168 to 0.171	0.3840
commentary_only	2023	time_after_s	token_within_player	exploratory	163	0.185	-0.156 to 0.156	0.0179
commentary_only	2023	time_after_s	utterance	exploratory	163	0.185	-0.162 to 0.167	0.0259
all_tokens	2023	dead_time_before_s	token_within_player	sensitivity (pre-	184	0.024	-0.157 to 0.127	0.7564

references	match	time	permutation	status	tokens	rho	null 95%	p
				registered token set)				
all_tokens	2023	dead_time_before_s	utterance	exploratory	184	0.024	-0.154 to 0.153	0.7615
all_tokens	2023	time_after_s	token_within_player	exploratory	193	0.086	-0.143 to 0.140	0.2305
all_tokens	2023	time_after_s	utterance	exploratory	193	0.086	-0.151 to 0.154	0.2565

The surname is the commonest form in every player x slot stratum, so most cells hold one or two types whatever the context; with these counts the thrift test would detect only a strong context-binding (Table 4.3: theta = 0.30 for 2019, 0.42 for 2023). The context is a clip-level tercile, while a clip's text spans the rally and the following dead time, so the cells do not encode the time available at the moment a name is chosen.

7. Top-10 repeated n-grams and systems for the composition brief (Phase 3)

Pre-specified ranking: distinct utterances, after dropping items contained in a longer item with the same count. Revision 1: `official` = at least half of the occurrences fall inside the revised umpire/score-call patterns (`common.official_mask_v2`, section 1); the pre-registered S5 patterns never covered score calls, so the flag did not fire on them in v1. The pre-specified list is shown with the flag, then the commentary-only list. Situational slot: modal clip role, share of occurrences at pressure points (break/set/match/tie-break/deuce), median time after and before (s).

Table 7.1a. Pre-specified top 10, with the `official` flag.

rank	kind	item	utterances	occurrences	pool matches	fillers (top)	official	modal role	pressure %	median time after	medi dea tim befo
1	system	<NUM> 15	36	40	16	40:13; 30:12; love:7; 13:3	True	rally	3	25.080	31.22
2	system	15 <NUM>	29	30	16	love:8; 0:7; 30:6; 13:5	True	rally	3	25.440	32.20
3	formula	the first	24	28	18		False	rally	29	40.400	31.12
4	system	<NUM> 30	24	26	16	40:10; 15:6; 14:5; 0:3	True	rally	8	29.080	29.14
5	system	40 <NUM>	22	28	17	15:13; 30:10; love:4; 40:1	True	rally	14	29.620	30.04
6	formula	a little	22	24	18		False	rally	18	31.920	27.36
7	system	the <_> set	21	27	17	first:10; fifth:7; second:6; fourth:2	False	rally	14	36.240	32.64
8	system	<NAME> djokovic	19	21	6	novak:16; federer:3; roger:1; djokovic:1	False	rally	26	43.480	26.58
9	system	30 <NUM>	19	19	16	15:12; love:4; 40:2; 0:1	True	rally	0	29.440	22.14
10	formula	the match	17	17	15		False	rally	18	25.000	25.40

Table 7.1b. Pre-specified top 10, umpire/score-call items excluded (commentary only).

rank	kind	item	utterances	occurrences	pool matches	fillers (top)	official	modal role	pressure %	median time after	me de ti be
1	formula	the first	24	28	18		False	rally	29	40.400	31.
2	formula	a little	22	24	18		False	rally	18	31.920	27.
3	system	the <_> set	21	27	17	first:10; fifth:7; second:6; fourth:2	False	rally	14	36.240	32.
4	system	<NAME> djokovic	19	21	6	novak:16; federer:3; roger:1; djokovic:1	False	rally	26	43.480	26.
5	formula	the match	17	17	15		False	rally	18	25.000	25.
6	formula	from djokovic	16	16	4		False	rally	31	29.740	31.
7	formula	you know	15	18	16		False	rally	20	31.800	27.
8	formula	going to	15	17	18		False	rally	13	28.520	20.
9	system	novak <NAME>	15	17	7	djokovic:16; djokovic's:1	False	rally	27	45.200	26.
10	formula	novak djokovic	14	16	6		False	rally	21	46.380	26.

Table 7.2a. Exploratory top 10 restricted to n >= 3 repeats and systems with >= 2 fixed tokens, with the `official` flag.

rank	kind	item	utterances	occurrences	pool matches	fillers (top)	official	modal role	pressure %	median time after	med de tir bef
1	system	the <_> set	21	27	17	first:10; fifth:7; second:6; fourth:2	False	rally	14	36.240	32.6
2	formula	a little bit	14	15	18		False	rally	14	25.080	29.2
3	system	djokovic <_> the	11	11	4	finding:2; had:2; down:1; certainly:1	False	rally	27	27.360	25.0
4	system	federer <_> to	10	10	0	looking:3; going:1; trying:1; able:1	False	rally	20	44.760	50.7
5	system	by <_> games to	9	9	7	two:3; four:2; six:1; five:1	True	rally	22	91.640	19.7
6	system	in the <_> set	9	9	15	fifth:4; first:2; second:2; final:1	False	rally	22	41.360	33.1
7	system	in <_> fifth	9	9	4	the:6; this:3	False	rally	44	40.280	33.6
8	formula	the first set	8	10	14		False	rally	0	36.200	26.0
9	system	djokovic <_> to	8	8	3	trying:2; able:2; needs:1; serving:1	False	rally	38	40.680	49.4
10	system	federer <_> the	8	8	2	with:1; got:1;	False	rally	38	41.360	29.7

rank	kind	item	utterances	occurrences	pool matches	fillers (top)	official	modal role	pressure %	median time after	med de tir bef
						resisting:1; gets:1					

Table 7.2b. Exploratory top 10 restricted to n >= 3 repeats and systems with >= 2 fixed tokens, umpire/score-call items excluded (commentary only).

rank	kind	item	utterances	occurrences	pool matches	fillers (top)	official	modal role	pressure %	median time after	med de tir bef
1	system	the <_> set	21	27	17	first:10; fifth:7; second:6; fourth:2	False	rally	14	36.240	32.6
2	formula	a little bit	14	15	18		False	rally	14	25.080	29.2
3	system	djokovic <_> the	11	11	4	finding:2; had:2; down:1; certainly:1	False	rally	27	27.360	25.0
4	system	federer <_> to	10	10	0	looking:3; going:1; trying:1; able:1	False	rally	20	44.760	50.7
5	system	in the <_> set	9	9	15	fifth:4; first:2; second:2; final:1	False	rally	22	41.360	33.1
6	system	in <_> fifth	9	9	4	the:6; this:3	False	rally	44	40.280	33.6
7	formula	the first set	8	10	14		False	rally	0	36.200	26.0
8	system	djokovic <_> to	8	8	3	trying:2; able:2; needs:1; serving:1	False	rally	38	40.680	49.4
9	system	federer <_> the	8	8	2	with:1; got:1; resisting:1; gets:1	False	rally	38	41.360	29.7
10	system	the <_> serve	8	8	17	first:3; great:1; federer:1; body:1	False	rally	12	28.860	45.3

8. Kuiper

- Kuiper, Koenraad and Douglas Haggo. 1984. Livestock auctions, oral poetry, and ordinary language. Language in Society 13(2): 205-234. doi:10.1017/S0047404500010381 Status: **verified**. Record: https://api.crossref.org/works?query.bibliographic=Livestock+auctions+oral+poetry+and+ordinary+language+Kuiper+Haggo (Crossref); https://www.cambridge.org/core/product/identifier/S0047404500010381/type/journal_article (publisher page, abstract). Confirms: authors, title, journal, volume, issue, pages, year; abstract: stock auctioneers "use an oral formulaic technique"; the technique "is a response to performance constraints which place a heavy load on short term memory"; "the difference between traditional oral formulaic and ordinary spoken language is one of degree, not kind".
- Kuiper, Koenraad. 1996. Smooth Talkers: The Linguistic Performance of Auctioneers and Sportscasters. Mahwah, NJ: Lawrence Erlbaum. xiv + 110 pp. Status: **verified (bibliographic only)**. Record: Crossref records of reviews: Coupland 1998, Language in Society, pp. 397-399, doi:10.1017/S0047404598243041 (title gives

"Mahwah, NJ: Erlbaum, 1996. Pp. xiv, 110"); Higgins 1997, Language, doi:10.2307/416088; Moser-Mercer 1996, Interpreting, pp. 267-270, doi:10.1075/intp.1.2.08mos. Confirms: title, subtitle, publisher, place, year, pagination (from the review titles); the book itself was not read, so nothing is claimed about its content.

- Kuiper, Koenraad. 2004. Formulaic performance in conventionalised varieties of speech. In: Formulaic Sequences (Language Learning and Language Teaching 9), 37-54. John Benjamins. doi:10.1075/lllt.9.04kui Status: **verified (bibliographic only)**. Record: https://api.crossref.org/works/10.1075/lllt.9.04kui. Confirms: author, chapter title, book title, series, pages, publisher, year; the volume editor is not in the record [unverified].
- Kuiper, Koenraad. 2009. Formulaic Genres. Palgrave Macmillan. doi:10.1057/9780230241657; chapter 2 "A Day at the Races", pp. 26-41, doi:10.1057/9780230241657_2 Status: **verified (bibliographic only)**. Record: https://api.crossref.org/works/10.1057/9780230241657_2 and https://api.crossref.org/works?query.author=Kuiper+Koenraad. Confirms: book and chapter titles, pages, publisher, year; chapter not read: that it treats race calling is [unverified].
- Kuiper, Koenraad and Austin [first name unverified]. 1990 [unverified]. Chapter on the speech of New Zealand race callers [title and pages unverified]. In A. Bell and J. Holmes (eds), New Zealand Ways of Speaking English. Clevedon: Multilingual Matters, 1990. Status: **[unverified]**. Record: Open Library edition records of the volume (https://openlibrary.org/books/OL2228004M.json: Multilingual Matters 1990, 305 pp.; OL1335680M: Victoria University Press 1990); Crossref record of Locker 1991 review (doi:10.5070/L421005133; title gives "Clevedon, UK: Multilingual Matters, 1990. 305 pp."). Confirms: only the edited volume (editors, publisher, place, year, 305 pp.) is confirmed; no fetched record lists a Kuiper and Austin chapter.

What the verified records say. Only the abstract of Kuiper and Haggo (1984) was read (record above): stock auctioneers "use an oral formulaic technique"; the technique "is a response to performance constraints which place a heavy load on short term memory"; and "the difference between traditional oral formulaic and ordinary spoken language is one of degree, not kind". For Kuiper (1996), Smooth Talkers, whose subtitle names auctioneers and sportscasters, and for Kuiper (2004, 2009) only bibliographic data were verified; nothing is claimed about their content, and the race-calling chapter attributed to Kuiper and Austin is [unverified].

What the data show.

- Repetition in the commentary is real and above a word-frequency floor: 44.2% of the 2019 final's tokens lie in n-grams repeated within the match, 19.4% in repeats of three or more words (shuffled floor 0.5%), and 24.4% of its tokens lie in n >= 3 repeats already found in the 18 other matches; 919 of the 1194 repeated types of the final recur in at least one other match (Tables K, 2.3, 3.1b).
- Degree: no auctioneer, race-call or other sportscaster corpus was measured with this procedure, so the commentary cannot be placed on a scale relative to Kuiper and Haggo's auctioneers. Among the corpora that were measured, the TV pool has lower matched-size split-half coverage than Cornell written live text and press answers under every definition in Table 3.2b; these are corpus contrasts (section 9.3), and the TV-vs-press comparison is indeterminate pending a WER estimate (C1/R1).
- Performance constraint: if repetition relieved time pressure, coverage and naming should shift toward the clips with the least time. No such shift was detected (C3a, C3b, C4, C5 and replications), but the tests could detect only large effects (Table 4.3: about 6-7 pp of coverage, theta = 0.30, rho = 0.17), and available time is assigned per clip rather than per phrase. The data therefore say nothing either way about whether tennis commentary carries the memory load described for auctioneers.

9. Limitations

1. **ASR and the TV-vs-press contrast (C1/R1).** The text is WhisperX output with no audio; the word error rate is unknown. The corpus's hand-read figure, 22 errors per 1,000 words (Wilson 95% 12-39, `corpus/reports/asr_proxy.json`), counts only visible errors and is a lower bound; plausible-word substitutions are invisible. Four things make the TV-vs-press contrast uninterpretable as a statement about commentary. (i) ASR noise alone can produce it: injected substitutions bring the held-out press value to the TV value at e* = 0.122 (split-half 0.142), a rate that cannot be excluded because only a lower bound on the TV error rate exists. (ii) Transcription convention: press answers are edited stenographic transcripts (disfluencies removed, sentences complete), while the TV text is raw ASR in clip windows with mid-sentence boundaries; both differences raise press repetition independently of how anyone speaks. (iii) The held-out designs are not analogous: for TV, I is other matches with mostly different commentators; for press, I is other interviewees of the same genre, transcription house and questioners. (iv) The v1 claim that the one-match TV measurement set favours TV is withdrawn: a concentrated M helps only if its recurrent items are in I, and 275 of the final's repeated types occur in no pool match. In the post hoc check D5(ii-b), where the press M comes from a median of 2 interviewees, press held-out coverage is 66.4% against 66.1% with M spread over many interviewees, so concentrating M did not visibly change the press value. The noise model is not a bound either: independent uniform substitutions drawn from the unigram distribution are neither clustered (real errors concentrate in names, score calls and overlapping speech) nor include deletions, insertions or word merges, and frequent-word replacements can create matches. Settling C1/R1 needs a measured WER on a sample with audio (FOR_HUMAN.md) or a within-convention comparison (e.g. press-conference audio through the same ASR pipeline).
2. **Clip-level text and time.** No word times: a clip's text spans the rally and the following dead time, so 'available time' is assigned per clip, not per phrase. C3a's predictor `dead_time_before_s` is the interval before the point, when the clip's text has not started, so it is misaligned with the text; C3b (`time_after_s`) is the apt predictor. The label permutations of C3-C5 treat adjacent clips as exchangeable although coverage and context may be serially correlated; with circular-shift nulls instead (Table 4.4) the p-values range from 0.2070 to 0.9033, so no conclusion changes. The phase tags are text-based heuristics; `between_points` clips are score calls by definition.
3. **Corpus contrasts, not medium contrasts.** TV pool, Cornell and press differ in matches (Cornell has no match ids), outlet (one live-text outlet with a house style), period (press record dates 2007-2015, TV 2019-2025), transcription (raw ASR vs edited prose vs edited stenography) and segmentation (clip windows vs updates vs answers); matched size equalises tokens only. A same-match comparison (TV and written minute-by-minute of one match, FOR_HUMAN.md) would be needed before 'medium' is claimed.
4. **What the measure is.** Repeated-n-gram coverage rises with vocabulary concentration and genre templating, so it orders texts by repetition rate, not by oral-formulaic composition; the stricter family (Tables K, 3.1b, 3.2b) shows how much rests on short repeats. The shuffled-word baseline destroys syntax and is a floor, not a null that preserves local grammar.
5. **One broadcaster per match, no speaker labels.** Commentator identity, broadcaster and match are confounded in every per-stream comparison; the umpire's and Hawk-Eye's voices are inside the text (S5/S5b mask the clearest calls; score calls are said by both).
6. **De-duplication.** Exact repeats of the previous clip's text (within 3 clips) were removed by the corpus builder; shorter overlaps remain. S4 (repeats at least 4 clips apart) and S1 (`text_dedup`) bound the effect.
7. **Size dependence.** Coverage rises with the size of the identification set; only size-matched comparisons are interpretable (Tables 3.2-3.4). For Cornell the pre-registered held-out I could not reach 100,000 tokens (median 65,444); D5(ii-b) shows the effect of a larger I with a differently drawn M.
8. **Referring expressions.** The epithet inventory is a fixed regex family checked against a discovery list; referents of descriptive epithets and the slot validation are single-annotator hand verdicts; pronouns are not resolved. Umpire patterns are excluded by rule, which may miss unusual calls or catch a commentator echoing one. The thrift and extension tests have little power (Table 4.3).

9. **CIs** condition on the repeated-n-gram inventory; split-half and random-split distributions (Table 3.7) show the identification variability. Utterances differ in length across corpora (mean tokens in section 0); because a repeat must recur in two distinct utterances, the shorter TV utterances make that criterion, if anything, easier for TV to meet.

10. Revision log (after `review/critic_analysis_v1.md`; details in plan.md addendum 2)

- A1: C1/R1 relabelled 'indeterminate pending a WER estimate'; the 'asymmetry favours TV' claim withdrawn; crossing points e* computed (f04b, held-out R raised to 50); the substantive discussion moved to section 9.1.
- B1: C1/R1/C2 use the 95% interval of replicate differences; p shown with its floor; D5(i) R raised from 200 to 1000 (f04 total runtime 17.5 min on 4 CPUs; D5(i) stage 6.3 min).
- B2: Cornell held-out I size corrected (median 65,444, range 56,574-76,850 tokens, not 103,675); deviation logged; D5(ii-b) added (Table 3.2).
- B3: measure renamed repeated-n-gram coverage; stricter family (n >= 3, n >= 4, content-only, content n >= 3) with bootstrap CIs for in-sample, split-half, pool -> 2019, 2019 -> 2023 and pool -> 2023, and for the corpus contrasts; shuffled baselines and excess for every definition (Tables K, 3.1b, 3.2b, 3.2c).
- B4: Kuiper section limited to verified records and data; minimum detectable effects simulated for C3-C5 (Table 4.3).
- B6: umpire-pattern references excluded from C4/C5/R4/R5 (rules in section 1); all-reference results kept as sensitivity (Tables 4.1b, 6.1b).
- B7: every TV/Cornell/press comparison labelled a corpus contrast.
- Minor: Duggan record fetched (criterion wording [unverified]); dates and replicate counts in the text read from data; top-10 `official` flag fires on score calls and `thank you`; top-filler share of systems shown; C3a interval and serial correlation noted, circular-shift check added (Table 4.4); noise-table caption explains the different meaning of the TV held-out range.

11. Files

`plan.md` (pre-specification and addenda), `common.py` (tokeniser, STOP, numerals, identification, coverage family, masks, contexts), `refexpr.py`, scripts `f01-f09` (`f04b` and `f09` post hoc), `make_report.py`, `run_all.sh`; hand inputs in `hand/` (`duggan_verification.tsv`, `kuiper_verification.tsv`, `epithet_referents.tsv`, `slot_sample40.tsv`); all tables in `results/` (`coverage_family.csv`, `density_*.csv` (column `density` = repeated-n-gram coverage), `medium_replicates.csv`, `confirmatory.csv`, `medium_contrasts.csv`, `power_curves.csv`, `mde_summary.json`, `circular_shift_tests.json`, `asr_noise_*`, `refexpr_*`, `thrift_*`, `formulas_2019.tsv`, `systems_2019.tsv`, `top10_for_brief*.csv`, `deviations.json`). Transcript excerpts in committed files are at most 12 words (repeated n-grams 915 words in total, systems 435) or 15 words (hand files, epithet candidates).